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# Adjoint Inversion Reveals Holographic Superposition and Destructive Interference in CNN Classifiers

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## Abstract

A foundational assumption of CNN interpretability, that deep encoder stages physically suppress non-discriminative pixels and that the classifier acts as a selector over an already-cleaned feature pool, has remained empirically untested at the pixel level, because no existing visualization tool offers an algorithmic guarantee against fabricating spatial structure. We close this gap on both fronts. We introduce a hallucination-free inversion framework built on magnitude and phase decoupling and a cascade of Local Adjoint Correctors deployed precisely at Markovian down-sampling boundaries; an algebraic identity guarantees that the spatial gradient support of every reconstruction is contained in the effective receptive fields of genuinely active channels. Used as a geometric probe, the framework yields, the first pixel-level direct evidence of strong superposition in vision encoders: per-channel inversions are uniformly holographic, and positive and negative weight reconstructions of any class are visually and energetically indistinguishable, yet their algebraic sum sharply concentrates on the foreground. This signature is precisely predicted by, and only by, an account in which classification is implemented as destructive interference: the classifier’s weights cancel a shared background direction in pixel space and constructively assemble class discriminative residuals, directly falsifying the Spatial Funnel Hypothesis. The interference account identifies a precise geometric quantity, the volume of the admissible interference subspace, that governs how many channels a classifier needs to operate; we prove this volume is dual to the GAP covariance determinant, yielding a covariance volume channel selection algorithm with a  $(1 - 1/e)$  approximation guarantee. The algorithm renders out of distribution failure as a measurable collapse of the very covariance volume that interference based classification requires.

## 1 Introduction

For more than a decade, tools for understanding what convolutional neural networks (CNNs) compute have been dominated by saliency and attribution methods, including Grad-CAM and its successors [26, 2, 28] and perturbation based approaches [24, 7]. The visual legibility of their outputs has shaped the community’s intuition about CNN decisions [8, 30]: deep convolutional stages are believed to progressively suppress non-discriminative pixels, concentrating energy on a localized foreground, and the final classifier is believed to select among an already cleaned feature pool. We refer to this picture as the Spatial Funnel Hypothesis (SFH). Although no individual paper commits to SFH explicitly, it is the implicit physical reading that makes existing visualizations intuitively legible, and it has never been tested at the pixel level. Verifying SFH at the pixel level requires two properties that are not jointly satisfied by existing visualization frameworks: (i) algorithmic faithfulness, i.e., a guarantee grounded in algebraic identity that the visualization does not introduce spatial structure absent from the encoder’s response, and (ii) compositional access, i.e., the ability to

isolate and compose individual channels and classifier directions in pixel space. Existing inversion methods [18, 4] rely on optimization with natural image priors and do not provide such guarantees, while generative approaches [20] can synthesize plausible spatial structure beyond that directly encoded in the representation. Meanwhile, CAM-style methods produce class-specific heatmaps but lack explicit channel-level control, whereas per-channel feature visualization methods [22] yield highly optimized yet synthetic patterns that are not grounded in any particular input instance.

We close this gap with an inversion framework built on two algebraic ideas. The first is a magnitude and phase decoupling: the absolute semantic importance of each channel is distilled, on the forward path, into a layer wise simplex measure  $E_{l,c}$ , while the spatial contour of the same concept is recovered, on the backward path, as a scale free spatial basis  $\tilde{V}_{l,c}$  via the channel selective Vector Jacobian Product. The second is the Local Adjoint Corrector (LAC), a Markovian transition guard deployed precisely at the manifold boundaries where the discrete adjoint signal is structurally shattered by zero insertion upsampling. Per channel GroupNorm[29] inside each LAC algebraically annihilates the dominant noise carriers without smoothing the spatial phase. The synthesis  $\hat{X}_l = \sum_c E_{l,c} \tilde{V}_{l,c}$  recovers a pixel space reconstruction whose spatial gradient support is, by an algebraic identity, strictly contained in the effective receptive fields of genuinely active channels (Corollary 1).

Used as a geometric probe, the framework reveals a structure of CNN classification sharply at odds with SFH and constituting the first pixel-level direct evidence of strong superposition in vision encoders, previously inferred only from activation statistics in language models [5]. Three signatures, independently incompatible with SFH, emerge consistently across architectures. First, individual deep channels recover a holographic rendering of the input scene, preserving foreground, background, and contextual layout. Second, for any class direction, positive and negative weight pixel space reconstructions are visually indistinguishable and exhibit identical foreground energy ratios, yet their algebraic sum exhibits sharp foreground concentration. Third, breaking the circular reasoning of sparse autoencoders[1] (which presuppose the sparse basis they aim to confirm), rank-one structure of per-image inversion Gram matrix captures  $> 20\%$  of total energy in a single shared direction.

These signatures are precisely and only predicted by a different mechanistic account we formulate and verify: classification is implemented as destructive interference in the pixel manifold. Each channel inversion decomposes as  $\tilde{V}_{L-1,i}(X) = C_i(X)B(X) + \delta_i(X)$ , where  $B(X)$  is a shared, low rank, high energy background direction,  $C_i(X) \in \mathbb{R}$  is the scalar projection coefficient of channel  $i$  onto that direction and  $\delta_i(X)$  is a channel idiosyncratic residual. By the linearity of the adjoint operator, the classifier’s scalar weights  $w^{(c)}$  in logit space are algebraically equivalent to a geometric superposition of these inversions in pixel space; a well-trained classifier’s weights simultaneously cancel the shared background and constructively align the residual sum on the class discriminative object. The classifier is not a selector but a precisely tuned linear interferometer. The interference account identifies a precise geometric quantity, the volume of the admissible interference subspace in pixel space, governing how many channels it needs. We prove an exact duality (Theorem 2) between this volume and the determinant of the GAP covariance matrix, yielding a greedy channel selection algorithm with a  $(1 - 1/e)$  approximation guarantee[19]. It preserves accuracy under aggressive compression, withstands corruption and cross dataset transfer, and renders out of distribution failure as a measurable collapse of the very covariance volume that interference based classification requires.

Contributions: (i) hallucination-free inversion framework with algebraic theorems for spatial fidelity, zero structural hallucination, and path invariance; (ii) first pixel-level evidence of strong superposition in vision encoders, falsifying both SFH structural predictions; (iii) algebraic reinterpretation of CNN classification as destructive interference in the pixel manifold

## 2 Method

### 2.1 Why Raw VJP Fails: Two Algebraic Pathologies

**Crisis I (zero insertion shock).** Modern CNNs reduce spatial resolution by stride  $s \geq 2$  at stage boundaries; in the forward pass this acts as decimation, but the chain rule forces the corresponding adjoint to be zero insertion upsampling  $\uparrow_s$ . When a VJP traverses a stage boundary on its way back to the pixel space, its spatial continuity is structurally shattered into a “nail bed” of high frequency sampling spikes(checkerboard artifacts)[21]. The rigorous derivation, which formalises this pathology as a discrete Dirac comb modulating the gradient continuum, is given in Supplementary A.1.

91 **Crisis II (magnitude phase entanglement).** The raw VJP rigidly couples two physically distinct  
 92 quantities in the same vector: the semantic importance of a concept, encoded by the magnitude  
 93  $\|v\|$ , and its spatial contour, encoded by the unit direction  $v/\|v\|$ . This coupling leaves the phase  
 94 defenseless against additive noise. The two crises are entangled because the zero insertion spikes of  
 95 Crisis I act precisely as the noise that exploits the vulnerability of Crisis II. Concretely, when the  
 96 noise dominates the orthogonal direction, the angular deviation admits a strictly positive lower bound:

$$\left\| \frac{v + \epsilon}{\|v + \epsilon\|} - \frac{v}{\|v\|} \right\| \geq c = \frac{1}{\sqrt{(1+K)^2 + 1}} > 0, \text{ for } \|\epsilon_{\perp}\| \geq \|v\|, \|\epsilon_{\parallel}\| \leq K\|v\|, \quad (1)$$

97 where  $\epsilon_{\parallel}$  and  $\epsilon_{\perp}$  are the components of  $\epsilon$  parallel and orthogonal to  $v$ . The proof and the physical  
 98 justification of the parallel bound  $\|\epsilon_{\parallel}\| \leq K\|v\|$  are deferred to Supplementary A.1. Any direct  
 99 readout of the raw VJP is therefore mathematically flawed, and our framework is designed to  
 100 neutralise both pathologies by algebraic construction rather than by post hoc smoothing.

## 101 2.2 Forward Path and Backward Path

102 Decoupling magnitude and phase, we formalise the pretrained encoder as a *Learned Analysis Operator*  
 103  $\mathcal{A} : X \mapsto \{h_{l,c}\}_{l,c}$ . Unlike hand-crafted bases (e.g., Fourier, Wavelets), this operator unfolds the  
 104 image into a data-driven frame of semantic probes. The forward path distills each channel’s absolute  
 105 magnitude into a scalar, ignoring spatial topology. For a frozen encoder on  $X$ , layer  $l$  per-channel  
 106 activation is extracted via global average pooling and layer-wise  $L_1$  normalization:

$$Z_{l,c} = \text{GAP}(|h_{l,c}|) = \frac{1}{H_l W_l} \sum_{r,s} |h_{l,c}(r,s)|, \quad E_{l,c} = \frac{Z_{l,c}}{\sum_{c'} Z_{l,c'}}. \quad (2)$$

107 Two properties follow by construction. First, absolute scale defines importance, while polarity  
 108 (excitation or suppression) is left as a structural property of the spatial phase. Second,  $\sum_c E_{l,c} \equiv 1$   
 109 as an algebraic identity rather than a soft regulariser; the family  $\{E_{l,c}\}_c$  resides on the standard  
 110  $(C_l - 1)$ -simplex  $\Delta^{C_l-1}$  and acts as the encoder’s confidence distribution over semantic concepts at  
 111 layer  $l$ . Existence and non-degeneracy of the measure are routine and deferred to Supplementary A.2.

112 Having distilled magnitude on the forward path, we recover the spatial phase of each semantic concept  
 113 via a channel-selective VJP. For layer  $l$  channel  $c$ , we define the seed by zeroing all other channels:

$$\text{seed}_{l,c} = e_c \odot h_l, \quad \text{VJP}_{l,c} = (\partial h_l / \partial X)^T \text{seed}_{l,c}. \quad (3)$$

114 A short chain rule calculation (Supplementary B.2) shows that this seed produces the exact identity

$$\text{VJP}_{l,c} = \nabla_X \Phi_{l,c}(X), \quad \Phi_{l,c}(X) := \frac{1}{2} \|h_{l,c}(X)\|^2, \quad (4)$$

115 This identity reveals that the VJP is an *endogenous activation-weighted projection*: spatial positions  
 116 with stronger activations contribute proportionally more to the pixel-space gradient. Consequently,  
 117 the spatial distribution of the VJP strictly and autonomously aligns with the highly discriminative  
 118 receptive field actually attended by the encoder, without heuristic approximations or external priors.

119 Crucially, the global adjoint  $(\partial h_l / \partial X)^T$  is not a single smooth map. By the chain rule it factorises  
 120 into a strict Markovian sequence of stage-wise pullbacks,

$$(\partial h_l / \partial X)^T = (\partial h_{\text{stem}} / \partial X)^T \circ (\partial h_0 / \partial h_{\text{stem}})^T \circ \dots \circ (\partial h_l / \partial h_{l-1})^T, \quad (5)$$

121 where each factor crosses exactly one manifold boundary. Each such crossing is where the zero  
 122 insertion shock of Crisis I strikes. This decomposition therefore identifies, by the algebraic structure  
 123 of the chain rule itself, the exact set of locations where any corrective mechanism must be inserted.

## 124 2.3 LAC: Amplitude Stripping at Manifold Boundaries

125 We deploy a Local Adjoint Corrector (LAC) at every Markovian boundary identified above. Intra-  
 126 stage convolutions are not intervened upon, since they are isomorphic mappings on the same semantic  
 127 manifold whose adjoint flow is already smooth. Each LAC is a per channel GroupNorm with affine  
 128 parameters, configured so that each channel forms its own normalisation group (`num_groups` =  
 129 `num_channels`). Letting  $v \in \mathbb{R}^{H_l W_l}$  denote the flattened VJP for channel  $c$  and  $P_{\perp 1} = I - \frac{1}{H_l W_l} \mathbf{1}\mathbf{1}^T$   
 130 the orthogonal projector onto the zero-mean hyperplane, the LAC implements

$$\text{GN}(v) = \gamma_c \cdot \sqrt{H_l W_l} \cdot \frac{P_{\perp 1} v}{\|P_{\perp 1} v\|} + \beta_c. \quad (6)$$

131 This factorisation makes LAC’s actions explicit:  $P_{\perp 1}$  removes the DC drift carrying Crisis I structural  
 132 quantisation noise; division by  $\|P_{\perp 1} v\|$  strips the absolute magnitude coupling Crisis II; direction  
 133  $P_{\perp 1} v / \|P_{\perp 1} v\|$  preserves spatial phase. Crisis I noise carriers are thus algebraically annihilated in  
 134 the inner product space, not heuristically smoothed (Supplementary A.3 and B.3).

135 Composing the per stage LACs with the Jacobian transposes of (5) yields the cascade

$$\tilde{V}_{l,c} = \Psi_{\text{LAC}_{\text{stem}}} \circ J_{\text{stem}}^T \circ \Psi_{\text{LAC}_0} \circ J_0^T \circ \cdots \circ \Psi_{\text{LAC}_l} \circ J_l^T (\text{seed}_{l,c}), \quad (7)$$

136 whose output  $\tilde{V}_{l,c}$  is a dimensionless, scale-free basis: it carries only the concept’s spatial contour.  
 137 Magnitude-phase decoupling is now complete, with  $E_{l,c}$  encoding energy and  $\tilde{V}_{l,c}$  encoding structure.

## 138 2.4 Synthesis, Training, and Inference

139 With the energy measure  $E_{l,c}$  distilled in the forward path and the scale-free spatial basis  $\tilde{V}_{l,c}$  refined  
 140 in the backward adjoint path. For each layer  $l \in \{0, \dots, L-1\}$ , we synthesise the pixel-space  
 141 reconstruction as a layer-wise weighted superposition:

$$\hat{X}_l = \sum_{c=1}^{C_l} E_{l,c} \cdot \tilde{V}_{l,c}, \quad (8)$$

142 where weights are the forward simplex measure and bases derive from the same layer’s adjoint. Layer-  
 143 wise restriction ensures geometric commensurability:  $E_{l,c}$  and  $\tilde{V}_{l,c}$  share an abstraction level, making  
 144 their combination well-defined; cross-stage mixing is not. Since (8) is strictly additive, isolating the  
 145  $c$ -th term yields a per-channel pixel-space visualisation  $\hat{X}_{l,c} = E_{l,c} \tilde{V}_{l,c}$  without retraining.

146 The framework is trained on a single objective, the deepest stage reconstruction

$$\mathcal{L} = \|X - \hat{X}_{L-1}\|_1. \quad (9)$$

147 By (7), the adjoint path from stage  $L-1$  traverses every LAC in the system, so this single path  
 148 provides full gradient coverage of all trainable parameters. Detailed justifications for this single-path  
 149 choice and for the  $L_1$  norm are given in Supplementary A.4. At inference, the trained LAC is reused  
 150 to produce all  $L$  layer-wise inversions  $\{\hat{X}_l\}_{l=0}^{L-1}$  without any retraining, since the LAC modules  
 151 required by stage  $l < L-1$  form a strict subset of those exercised by the deepest path.

## 152 2.5 Theoretical Guarantees (Zero Structural Hallucination)

153 **Theorem 1** (Spatial Fidelity, see Supp. B.4). *For a ReLU based encoder, define the effective receptive*  
 154 *field of channel  $c$  at layer  $l$  on input  $X$  as*

$$\text{EF}_{l,c}(X) = \{(i, j) : \exists (r, s) \text{ s.t. } h_{l,c}(r, s) > 0 \text{ and } \partial h_{l,c}(r, s) / \partial X(i, j) \neq 0\}.$$

$$\text{Then } \text{supp}(\text{VJP}_{l,c}) \subseteq \text{EF}_{l,c}(X). \quad (10)$$

155 VJP never assigns nonzero attribution to any pixel outside spatial region actively used by the encoder.

156 **Corollary 1** (Zero Structural Hallucination, see Supp. B.7). *Let  $\nabla_{i,j}$  denote the discrete spatial*  
 157 *gradient operator on the pixel grid, and  $\overline{(\cdot)}$  the one-pixel topological closure. Then*

$$\text{supp}(\nabla_{i,j} \hat{X}_l) \subseteq \overline{\bigcup_{c: E_{l,c} > 0} \text{EF}_{l,c}(X)}. \quad (11)$$

158 Corollary 1 is a strict algebraic identity, independent of empirical regularisation or the asymptotic  
 159 convergence of LAC’s affine parameters. Every synthesised spatial detail originates exclusively from  
 160 the effective receptive fields of active encoder channels. Lacking the degrees of freedom to fabricate  
 161 spatial structures, trainable LAC parameters render the framework mathematically hallucination-free.

162 Three additional theoretical results undergird the framework. (i) Lemma B.1 establishes the asymp-  
 163 totic convergence  $\beta^* \rightarrow 0$ ,  $\gamma^* > 0$  of LAC’s affine parameters under the  $L_1$  loss. (ii) Theorem B.5  
 164 and Corollary B.6 prove the algebraic equivalence (up to a tightly bounded cascade direction residual)  
 165 between  $\hat{X}_l$  and a weighted superposition of unit-normalised gradient-ascent feature visualisations,  
 166 providing a formal mathematical bridge to the feature visualisation literature. (iii) Proposition B.8  
 167 establishes the path invariance of shared LAC parameters across source stages, which underpins the  
 168 framework’s plug-and-play extension to arbitrary differentiable downstream heads (Section 3.5).

## 3 Experiments

### 3.1 Reconstruction Fidelity Against Inversion Baselines

We compare four baselines inverting the frozen encoder’s deepest features: **Vanilla Gradient** (raw VJP), **Mahendran & Vedaldi** [18] (TV/ $\ell_2$  opt.), **Robust Inversion** [6] (robust), and **UpConvNet** [4] (feedforward decoder). Table 1 reports SSIM ( $\uparrow$ ) and LPIPS ( $\downarrow$ ) across five datasets (ImageNet[25], CIFAR-100[14], CUB-200[27], Pets[23], Dogs[13]) and four backbones: *R-50/R-18/D-121/C-Base*.

Table 1: Quantitative comparison of feature inversion fidelity(%). Format *R-50/R-18/D-121/C-Base*.

| Method      | Metric             | ImageNet                   | CIFAR-100               | CUB-200                    | Pets                       | Dogs                       |
|-------------|--------------------|----------------------------|-------------------------|----------------------------|----------------------------|----------------------------|
| <b>Ours</b> | SSIM $\uparrow$    | <b>41.7/34.6/38.5/43.2</b> | 67.6/36.8/40.4/-        | <b>54.8/44.2/40.7/47.7</b> | <b>52.1/35.4/34.9/37.6</b> | <b>51.7/36.1/34.8/38.1</b> |
|             | LPIPS $\downarrow$ | <b>43.0/57.7/45.8/53.5</b> | 85.4/93.8/94.6/-        | <b>52.9/64.4/61.5/64.6</b> | <b>54.3/72.7/63.8/71.5</b> | <b>51.5/72.1/63.5/70.0</b> |
| Vanilla     | SSIM $\uparrow$    | 4.8/2.5/4.3/3.7            | 4.5/2.0/4.1/-           | 5.0/3.1/4.9/6.1            | 3.5/1.9/3.0/4.1            | 5.1/3.2/4.7/5.2            |
|             | LPIPS $\downarrow$ | 109.5/119.0/107.9/108.8    | 140.8/173.0/144.6/-     | 129.1/140.4/133.9/134.1    | 124.9/134.3/119.6/129.9    | 128.0/139.6/123.6/125.7    |
| Mahendran   | SSIM $\uparrow$    | 7.5/6.2/6.9/8.7            | 13.8/7.3/9.8/-          | 10.5/8.4/6.2/12.5          | 13.2/7.5/9.6/13.9          | 10.5/6.5/8.2/14.1          |
|             | LPIPS $\downarrow$ | 99.0/107.0/108.8/104.5     | 136.4/164.0/142.7/-     | 118.6/133.2/127.3/95.1     | 113.3/125.0/112.2/87.9     | 115.6/131.9/118.3/81.3     |
| Robust      | SSIM $\uparrow$    | 7.6/6.5/8.7/9.6            | 8.7/7.4/10.1/-          | 7.1/4.7/4.9/8.9            | 7.4/4.7/6.3/9.7            | 6.2/4.4/5.2/10.1           |
|             | LPIPS $\downarrow$ | 105.2/113.6/116.8/101.6    | 142.7/165.4/143.0/-     | 121.0/132.1/131.5/105.8    | 119.2/124.1/121.7/106.0    | 120.8/131.4/123.9/99.6     |
| UpConvNet   | SSIM $\uparrow$    | 15.9/18.5/14.5/12.0        | <b>69.8/63.4/67.4/-</b> | 43.5/36.4/38.5/36.2        | 43.5/39.6/41.9/43.1        | 36.4/32.3/33.6/37.4        |
|             | LPIPS $\downarrow$ | 89.3/92.1/92.2/89.0        | <b>62.8/73.8/75.4/-</b> | 64.1/76.8/71.2/87.0        | 76.5/82.6/77.9/76.0        | 84.4/91.4/87.0/85.2        |

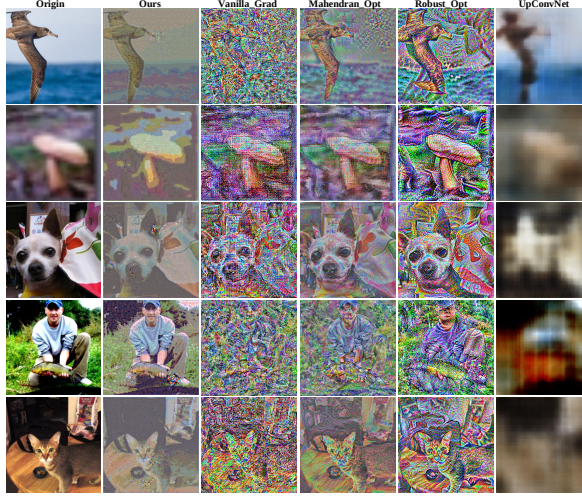


Figure 1: Qualitative inversion comparison; Our framework preserves structural details and sharp contours.

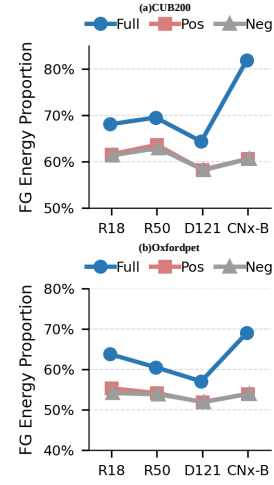


Figure 2: Foreground energy proportion FG.

Our framework dominates 4 of 5 datasets across all backbones. On ImageNet, it surpasses the strongest baseline (UpConvNet) by 2.1–3.6 $\times$  in SSIM and 35–50% in LPIPS, with similar margins on CUB-200, Pets, and Dogs. Vanilla Gradient fails (SSIM  $\leq$  6.1), confirming Section 2.1’s dual pathologies; optimization baselines remain low-fidelity, proving heuristic priors cannot resolve the algebraic crisis. The sole exception is CIFAR-100, where UpConvNet’s over-smoothed outputs coincidentally match the diffuse  $32 \times 32 \rightarrow 224 \times 224$  targets, penalizing our method for recovering the sharp features inherent to pre-training. Figure 1 corroborates this visually (details in Supp. C.1).

### 3.2 Holographic Superposition and Destructive Interference

#### 3.2.1 The Foreground Energy Paradox

**Class-directional inversion.** Let  $h_i(X) := \text{GAP}(\mathbf{h}_{L-1,i}(X))$  and define the importance weight  $w_i^{(c)} = \partial \log \text{it}_c(X) / \partial h_i(X)$ . By the linearity of the adjoint together with the additive synthesis of Section 2.4, the pixel-space image of the class direction is

$$\hat{X}^{(c)}(X) = \sum_{i=1}^{C_{L-1}} w_i^{(c)} \tilde{V}_{L-1,i}(X), \quad \hat{X}^{(c)} = \sum_{w_i^{(c)} > 0} w_i^{(c)} \tilde{V}_{L-1,i} + \sum_{w_i^{(c)} < 0} w_i^{(c)} \tilde{V}_{L-1,i}, \quad (12)$$



Figure 3: (a) Per-channel inversions  $\tilde{V}_{L-1,i}(X)$  by energy rank: channels recover the full scene, falsifying P1. (b) Stage-wise inversions  $\hat{X}_{0-3}$ , class reconstruction  $\hat{X}^{(c)}$ , and sign hemispheres  $\hat{X}_{\pm}^{(c)}$ : deep stages retain the full scene with indistinguishable hemispheres, falsifying P2.

where the identity  $\nabla_X \sum_i w_i^{(c)} h_i(X) \equiv \sum_i w_i^{(c)} \nabla_X h_i(X)$  is an exact consequence of the chain rule, making this decomposition an algebraic property of the classifier rather than a heuristic attribution.

**Foreground energy proportion.** Given reconstruction  $\hat{X}$  and ground-truth mask  $M_{fg}$ , define  $FG(\hat{X}) := \sum_{(i,j) \in M_{fg}} \hat{X}(i,j)^2 / \sum_{(i,j)} \hat{X}(i,j)^2 \in [0, 1]$ . Evaluating on **CUB-200-2011** [27] and **Oxford-IIIT Pet** [23] across four ImageNet-pretrained backbones (ResNet-18/50, DenseNet-121, ConvNeXt-Base) using the top-1 prediction  $c$ , Figure 2 reveals three unambiguous patterns consistent across all pairs: **(O1)** positive and negative hemispheres show indistinguishable foreground energy proportions; **(O2)** the full reconstruction  $\hat{X}^{(c)}$  exhibits substantially higher foreground energy than either hemisphere; **(O3)** this gap grows monotonically with architectural capacity.

**The paradox.** Jointly the observations form a striking algebraic signature: foreground concentration is generated by the summation itself, not inherited from either sign hemisphere. Any account in which the classifier acts as a spatial selector predicts the opposite pattern. The empirical signature  $FG(\hat{X}_+^{(c)}) \approx FG(\hat{X}_-^{(c)}) \ll FG(\hat{X}^{(c)})$  admits no such explanation.

**Visual Falsification of the SFH.** If SFH (Section 1) holds, two structural predictions follow: **(P1)** per-channel inversions  $\tilde{V}_{L-1,i}(X)$  should be spatially sparse, with channels qualitatively differing in what they render; **(P2)** the positive-weight reconstruction  $\hat{X}_+^{(c)}$  should already resemble the foreground while  $\hat{X}_-^{(c)}$  expresses a complementary pattern. Figure 3 unambiguously falsifies both.

**Falsification of P1** (Fig. 3a). Across all energy ranks, every channel recovers a visually complete, globally-supported scene. No channel specializes locally; variation manifests strictly as a global semantic “lens” (texture emphasis, chromatic weighting, edge polarity), not spatial support. **Falsification of P2** (Fig. 3b). Two findings emerge: (1) Background pixels are never discarded: even the deepest stage  $\hat{X}_3$  retains full spatial extent, shifting only style and frequency rather than spatial support. (2)  $\hat{X}_+^{(c)}$  and  $\hat{X}_-^{(c)}$  are visually nearly indistinguishable, yet their algebraic sum sharply concentrates on the foreground, the precise visual counterpart to O1–O2.

The background is never removed by the encoder, never removed by either sign group of classifier weights, yet vanishes in their algebraic sum. This is empirical signature of cancellation, not filtering.

### 3.2.2 The Destructive-Interference Account

**H1: Holographic Superposition.** Every deep channel inversion decomposes as

$$\tilde{V}_{L-1,i}(X) = C_i(X) \cdot B(X) + \delta_i(X), \quad \langle \delta_i(X), B(X) \rangle \approx 0, \quad (13)$$

where  $B(X)$  is a low-rank background direction shared by all channels,  $C_i(X) \in \mathbb{R}$  is channel  $i$ ’s scalar projection onto it, and  $\delta_i(X)$  is its idiosyncratic residual carrying object-discriminative structure. The encoder retains  $B(X)$  because any common-mode component cancels from the logit of every class regardless of magnitude; preserving it costs nothing and protects representational

generality [5]. Empirically (Supp. C.4), the per-image inversion Gram matrix’s leading eigenvector captures 20.9% of total energy; rank-one subtraction expands inter-channel variance by  $1.27 \times 10^4 \times$ , widening the cosine-similarity range from  $[0.18, 0.21]$  to  $[-0.07, 0.67]$ . Eq. (13) thus constitutes the first direct pixel-level confirmation of strong superposition in vision encoders.

**H2: Classification as Destructive Interference.** Substituting H1 into the class-directional reconstruction yields the exact decomposition

$$\hat{X}^{(c)}(X) = B(X) \underbrace{\sum_i w_i^{(c)} C_i(X)}_{\alpha^{(c)}(X)} + \underbrace{\sum_i w_i^{(c)} \delta_i(X)}_{\text{foreground residual sum}}. \quad (14)$$

A well-trained classifier is one whose weights satisfy two algebraic boundary conditions: **(B1)**  $\alpha^{(c)}(X) \approx 0$  (sign-mixed cancellation of the shared background), and **(B2)** the residual sum aligns constructively on the class- $c$  object. Classifier is not selector but precisely-tuned linear interferometer.

H1 immediately explains O1 (each hemisphere retains background at full amplitude), and H2 immediately explains the jump in O2 (full sum satisfies B1 and background is algebraically annihilated). Detailed resolution of paradox and additional empirical verification of H1 are in Supp. C.4.

### 3.2.3 From Theory to Algorithm: Covariance-Volume Channel Selection

H1 and H2 identify channels necessary for classification. Retaining a subset  $S \subseteq \{1, \dots, C_{L-1}\}$  ( $|S| = k$ ) yields the admissible interference subspace under cancellation constraint (B1):  $\mathcal{F}_S(X) = \{\sum_{i \in S} w_i \tilde{V}_{L-1,i}(X) \mid \sum_{i \in S} w_i C_i(X) = 0\} \subset \mathbb{R}^{3H_0W_0}$ . The geometric capacity of this  $(k-1)$ -dimensional subspace is measured by its parallelepiped volume  $\text{Vol}(\mathcal{F}_S(X))$  (Supp. C.5); larger volumes admit richer foreground patterns under sign-mixed background cancellation.

**Theorem 2** (Covariance-Volume Duality). *Under Hypothesis H1 and a generic non-degeneracy condition, the admissible interference volume and the GAP covariance determinant share the same maximizers up to a data-independent positive constant:*

$$\arg \max_{|S|=k} \mathbb{E}_X[\log \text{Vol}^2(\mathcal{F}_S(X))] = \arg \max_{|S|=k} \log \det(\Sigma_S), \quad (15)$$

where  $\Sigma_S$  is the centered GAP covariance submatrix indexed by  $S$ .

The proof (Supp. C.5) reduces volume to a Gram determinant via constraint hyperplane orthonormal restriction, linking expected residual Gram to GAP covariance via LAC normalization. Direct optimization in inversion space is numerically degenerate because the LAC enforces  $\|\tilde{V}_{L-1,i}\|_2 = \gamma_i^* \sqrt{H_0 W_0}$  to machine precision (Supp. C.5); GAP space inherits same maximizers without constraint.

Direct optimization  $S^* = \arg \max_{|S|=k} \det(\Sigma_S)$  is NP-hard in general [3]; however,  $\log \det$  is monotone submodular on PSD matrices [15], so Algorithm 1 (partial Cholesky with diagonal pivoting) inherits the classical  $(1 - 1/e)$  approximation guarantee [19].

### 3.2.4 Empirical Validation: Compression and the Geometry of OOD Failure

Using a frozen ConvNeXt-Base, we compare three final-layer selection methods: **CovVol** (Algorithm 1), **Energy** (top- $k$   $\ell_2$  activation energy)[16], and **Random** (50-seed mean $\pm$ std). For pruning ratio  $\rho$ , we train a linear probe on  $\lceil (1 - \rho) C_{L-1} \rceil$  retained channels. Evaluation covers **CIFAR-100** (ID), **CIFAR-100-C** [11] (OOD), **Stanford Dogs** [13], and **Oxford-IIIT Pet** (cross-dataset).

**Finding 1: Robustness plateau refutes spatial specialization.** Under  $\sim 90\%$  pruning, all methods preserve full-model accuracy within fractions of a percent (CIFAR-100:  $88.23 \rightarrow 87.50$ , Dogs:  $91.75 \rightarrow 91.36$ , Pets:  $94.39 \rightarrow 93.46$ , CIFAR-100-C:  $62.77 \rightarrow 61.23$ ). Random selection matching principled selectors is precise signature of holographic superposition (H1): channels are informationally redundant, allowing interference geometry reconstruction from nearly any large subset.

**Finding 2: In the tail, interference geometry reveals itself.** Methods separate at extreme compression ( $\rho \geq 95\%$ ). At 98% pruning, CovVol outperforms Energy and Random on CIFAR-100 (82.20% vs. 80.17% vs.  $80.72 \pm 0.46\%$ ) and CIFAR-100-C (51.96 vs. 48.98 vs.  $50.99 \pm 0.63$ ). The CovVol-vs-Energy gap is uniformly larger on OOD than ID, confirming covariance-volume selection preserves interference capacity over mere training accuracy.

### Algorithm 1 Covariance-Volume Channel Selection

- 1: **Input:** GAP feature matrix  $H \in \mathbb{R}^{N \times C}$ ; retention size  $k$ .
- 2: Compute centered covariance  $\Sigma \leftarrow \frac{1}{N}(H - \bar{H})^\top (H - \bar{H})$ ; initialize  $\tilde{\Sigma} \leftarrow \Sigma$ ,  $S \leftarrow \emptyset$ .
- 3: **for**  $t = 1, \dots, k$  **do**
- 4:  $i^* \leftarrow \arg \max_{i \notin S} \tilde{\Sigma}_{ii}$   
 $\triangleright$  largest residual variance
- 5:  $S \leftarrow S \cup \{i^*\}$ ;  
 $\tilde{\Sigma}_{:,i^*} \leftarrow \tilde{\Sigma}_{:,i^*} / \tilde{\Sigma}_{i^*,i^*}$
- 6: **end for**
- 7: **Output:**  $S$ .

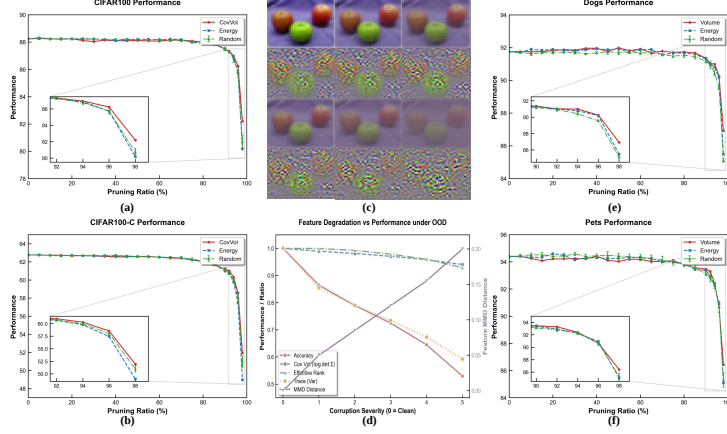


Figure 4: (a,b,e,f) Pruning ratio vs. accuracy on CIFAR-100 (ID), CIFAR-100-C (OOD), Stanford Dogs and Pets (cross-dataset). (c) Class reconstructions degrade under corruption. (d) Normalized accuracy,  $\log \det(\Sigma)$ , effective rank,  $\text{Tr}(\Sigma)$ , and MMD vs. corruption severity.

Table 2: Faithfulness (Insertion/Deletion AUC). Higher Ins / lower Del is better.

|             | CUB-200        |                  | Pet            |                  |
|-------------|----------------|------------------|----------------|------------------|
|             | Ins $\uparrow$ | Del $\downarrow$ | Ins $\uparrow$ | Del $\downarrow$ |
| G-CAM       | .563           | .489             | .746           | .641             |
| G-CAM++     | .628           | .409             | .698           | .654             |
| S-CAM       | .573           | .426             | .740           | .571             |
| <b>Ours</b> | <b>.762</b>    | <b>.164</b>      | <b>.817</b>    | <b>.384</b>      |

Table 3: Localization (mIoU/mAP/pAcc/Pointing). Best **bold**, second underlined.

|             | CUB-200     |             |             |             | Pet         |             |             |             |
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
|             | mIoU        | mAP         | pAcc        | Point       | mIoU        | mAP         | pAcc        | Point       |
| G-CAM       | .401        | .617        | .648        | .614        | .294        | .422        | .653        | .379        |
| G-CAM++     | .345        | .664        | .625        | .705        | .245        | .400        | .635        | .347        |
| S-CAM       | <b>.576</b> | .647        | <b>.698</b> | .544        | <b>.470</b> | <b>.482</b> | .619        | <b>.409</b> |
| <b>Ours</b> | .319        | <b>.748</b> | .573        | <b>.888</b> | <b>.428</b> | <b>.509</b> | <b>.698</b> | <b>.460</b> |

**Finding 3: OOD failure is interference breakdown.** Fig. 4(c) visualizes  $\hat{X}^{(c)}$  under increasing corruption: foreground dissolves and background bleeds in, visually signaling (B1)’s failure as  $B(X)$  is perturbed. Panel (d) plots normalized accuracy,  $\log \det(\Sigma)$ , effective rank,  $\text{Tr}(\Sigma)$ , and MMD [9] against severity  $s \in \{0, \dots, 5\}$ . All four geometric indicators of interference capacity decay monotonically, nearly proportionally to MMD growth. Notably, the clean effective rank ( $\sim 123$ , roughly 10% of 1024 channels) exactly explains Finding 1’s 90% robustness plateau: accuracy collapses only when the channel budget breaches the interference subspace’s intrinsic dimensionality. Out-of-distribution failure is not vague “distribution shift”; at the algebraic level, it is the erosion of the very covariance volume that interference-based classification requires.

### 3.3 Comparison with Gradient-Based Attribution

We compare official **Grad-CAM** [26], **Grad-CAM++** [2], and **Score-CAM** [28] on ConvNeXt-Base across CUB-200 and Oxford-IIIT Pet. Our saliency maps use the per-pixel  $\ell_2$  magnitude of  $\hat{X}^{(c)}$ .

**Faithfulness** (Table 2). Our framework dominates: on CUB-200, surpassing the strongest baseline by +13.3 Insertion and  $-24.6$  Deletion AUC (reaching 0.1641, under half the next-best 0.4096), with comparable +7.0 and  $-18.8$  gaps on Pet. This occurs because Grad-CAM baselines bilinearly interpolate coarse saliency, discarding high-frequency adjoint structures LAC recovers; conversely, Corollary 1 guarantees nonzero attributions lie within active receptive fields. **Localization** (Table 3). We win peak-localization metrics (mAP/Pointing on both, Pet pAcc) but yield CUB mIoU to Score-CAM. This trade-off is predictable: H2 dictates that sparse sub-object residuals  $\delta_i(X)$  drive classification. Attributions faithful to this mechanism concentrate on discriminative sub-regions over uniform silhouettes, penalizing bulk-overlap metrics. Score-CAM’s smoothed maps boost mIoU but lose mechanistic connection, evidenced by a Deletion AUC  $2.6\times$  ours (Supp. C.7).

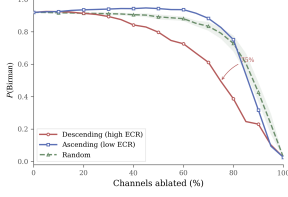


Figure 5: Channel ablation.

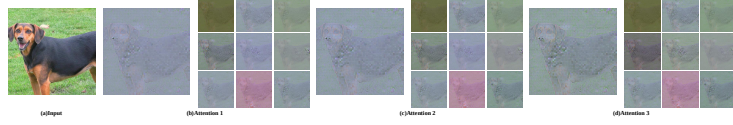


Figure 6: Layer-wise (large) and per-token ( $3 \times 3$ ) attention visualizations across three hybrid attention blocks.

### 3.4 Causal Concept Localization via Channel Ablation

If  $\delta_i(X)$  carries concept-specific structure (H1), ablating channels by concept-localization energy should causally damage the corresponding classification. For a reference  $X_{\text{ref}}$  and region  $\mathcal{R}$  (here, the tail of one Birman cat), we define the Energy Concentration Ratio  $\text{ECR}_i$  as the proportion of squared  $\ell_2$  inversion energy within  $\mathcal{R}$ :  $\text{ECR}_i := \sum_{(u,v) \in \mathcal{R}} \|\tilde{V}_i(u,v)\|^2 / \sum_{(u,v)} \|\tilde{V}_i(u,v)\|^2$ .

The shared background  $B(X_{\text{ref}})$  contributes a near-constant additive term across all channels (H1), so  $\text{ECR}_i$  variation is dominated by the residual  $\delta_i$ . We rank all channels by ECR using only  $X_{\text{ref}}$ , applying this ranking to every Birman test image under three ablation orders: **Descending** (high-ECR first), **Ascending** (low-ECR first), **Random** (50 seeds). Figure 5 reports softmax probability  $P(\text{Birman})$  vs ablation fraction. At 70% ablation, Descending collapses to 0.61, vs 0.83 for Random and 0.88 for Ascending; Ascending tracks above Random mid-ablation, confirming low-ECR channels are genuinely tail-irrelevant. However, beyond 80% ablation, Ascending drops below Random: retaining only highly homogeneous channels destroys the geometric diversity required to annihilate the shared background (H2). The separation  $\geq 0.3$  in  $P(\text{Birman})$  between Descending and Ascending is central: it would be impossible if channels were spatially unspecialized or sparsified onto object regions. To our knowledge, this is the first demonstration of causal concept localization at the channel level from a single spatial annotation, a capability unavailable to CAM-family methods.

### 3.5 Generalization to Attention-Based Heads

Because the LAC cascade trains solely on the encoder’s adjoint structure, it universally serves arbitrary differentiable downstream heads. We build a hybrid classifier using a frozen ConvNeXt-Base, flattening  $\mathbf{h}_{L-1}(X) \in \mathbb{R}^{1024 \times 7 \times 7}$  into 49 tokens processed by three trainable attention blocks  $\mathcal{A}_{1,2,3}$  and a linear head. Loading the unchanged LAC, any attention-pathway scalar  $\mathcal{Q}$  is visualized via  $\tilde{V}_{\mathcal{Q}}(X) = \Psi_{\text{LAC}_{\text{stem}}} \circ J_{\text{stem}}^T \circ \dots \circ \Psi_{\text{LAC}_{L-1}} \circ J_{L-1}^T (\partial \mathcal{Q} / \partial \mathbf{h}_{L-1})$ , directly feeding the head’s VJP through the LAC. We evaluate layer-wise  $\mathcal{Q}_\ell = \frac{1}{2} \|\mathcal{A}_\ell(\mathbf{T}(X))\|_F^2$  and per-token  $\mathcal{Q}_{\ell,t} = \frac{1}{2} \|\mathcal{A}_\ell(\mathbf{T}(X))_{t,:}\|_2^2$  queries. Fig. 6 confirms coherent layer aggregates and structurally specialized tokens (early tokens diffuse, deeper ones concentrate, adjacent ones neighbor). Unlike Grad-CAM, LAC’s single-training transfers all theoretical guarantees intact to new heads.

## 4 Conclusion

We introduce a hallucination-free inversion framework that algebraically guarantees zero hallucination through magnitude and phase decoupling alongside Local Adjoint Correctors at Markovian downsampling boundaries. Used geometrically, it provides the first pixel-level evidence of strong vision-encoder superposition and reveals classification as destructive interference in the pixel manifold. Specifically, classifier weights cancel shared backgrounds to assemble class-discriminative residuals, falsifying both structural predictions of the Spatial Funnel Hypothesis. Furthermore, this interference account yields a  $(1 - 1/e)$ -guaranteed covariance volume channel selection algorithm, enabling aggressive compression while exposing out-of-distribution failure as measurable covariance volume collapse. Although normalization may attenuate spatial contrast in sparse responses, our framework seamlessly visualizes downstream attention heads via the CNN’s adjoint. This success points to theoretical blueprint for future work on pure ViT backbones: adapting LAC principles to algebraically neutralize softmax-induced mixing noise and rescue relative phase trends. Ultimately, this advances mechanistic vision interpretability from heuristic attribution to exact algebraic geometry.

## References

- [1] Trenton Bricken, Adly Templeton, Joshua Batson, Brian Chen, Adam Jermyn, Tom Conerly, Nick Turner, Cem Anil, Carson Denison, Amanda Askill, et al. Towards monosemanticity: Decomposing language models with dictionary learning. *Transformer Circuits Thread*, 2(5):6, 2023.
- [2] Aditya Chattopadhyay, Anirban Sarkar, Prantik Howlader, and Vineeth N Balasubramanian. Grad-cam++: Generalized gradient-based visual explanations for deep convolutional networks. In *2018 IEEE winter conference on applications of computer vision (WACV)*, pages 839–847. IEEE, 2018.
- [3] Ali Civril and Malik Magdon-Ismael. On selecting a maximum volume sub-matrix of a matrix and related problems. *Theoretical Computer Science*, 410(47-49):4801–4811, 2009.
- [4] Alexey Dosovitskiy and Thomas Brox. Inverting visual representations with convolutional networks. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 4829–4837, 2016.
- [5] Nelson Elhage, Tristan Hume, Catherine Olsson, Nicholas Schiefer, Tom Henighan, Shauna Kravec, Zac Hatfield-Dodds, Robert Lasenby, Dawn Drain, Carol Chen, et al. Toy models of superposition. *arXiv preprint arXiv:2209.10652*, 2022.
- [6] Logan Engstrom, Andrew Ilyas, Shibani Santurkar, Dimitris Tsipras, Brandon Tran, and Aleksander Madry. Adversarial robustness as a prior for learned representations. *arXiv preprint arXiv:1906.00945*, 2019.
- [7] Ruth C Fong and Andrea Vedaldi. Interpretable explanations of black boxes by meaningful perturbation. In *Proceedings of the IEEE international conference on computer vision*, pages 3429–3437, 2017.
- [8] Ian Goodfellow, Yoshua Bengio, Aaron Courville, and Yoshua Bengio. *Deep learning*, volume 1. MIT press Cambridge, 2016.
- [9] Arthur Gretton, Karsten M Borgwardt, Malte J Rasch, Bernhard Schölkopf, and Alexander Smola. A kernel two-sample test. *The journal of machine learning research*, 13(1):723–773, 2012.
- [10] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 770–778, 2016.
- [11] Dan Hendrycks and Thomas Dietterich. Benchmarking neural network robustness to common corruptions and perturbations. *arXiv preprint arXiv:1903.12261*, 2019.
- [12] Gao Huang, Zhuang Liu, Laurens Van Der Maaten, and Kilian Q Weinberger. Densely connected convolutional networks. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 4700–4708, 2017.
- [13] Aditya Khosla, Nityananda Jayadevaprakash, Bangpeng Yao, and Fei-Fei Li. Novel dataset for fine-grained image categorization: Stanford dogs. In *Proc. CVPR workshop on fine-grained visual categorization (FGVC)*, volume 2, 2011.
- [14] Alex Krizhevsky, Geoffrey Hinton, et al. Learning multiple layers of features from tiny images. 2009.
- [15] Alex Kulesza and Ben Taskar. Determinantal point processes for machine learning. *Foundations and Trends® in Machine Learning*, 5(2-3):123–286, 2012.
- [16] Hao Li, Asim Kadav, Igor Durdanovic, Hanan Samet, and Hans Peter Graf. Pruning filters for efficient convnets. *arXiv preprint arXiv:1608.08710*, 2016.
- [17] Zhuang Liu, Hanzi Mao, Chao-Yuan Wu, Christoph Feichtenhofer, Trevor Darrell, and Saining Xie. A convnet for the 2020s. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 11976–11986, 2022.

- [18] Aravindh Mahendran and Andrea Vedaldi. Understanding deep image representations by inverting them. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 5188–5196, 2015.
- [19] George L Nemhauser, Laurence A Wolsey, and Marshall L Fisher. An analysis of approximations for maximizing submodular set functions—i. *Mathematical programming*, 14(1):265–294, 1978.
- [20] Anh Nguyen, Jeff Clune, Yoshua Bengio, Alexey Dosovitskiy, and Jason Yosinski. Plug & play generative networks: Conditional iterative generation of images in latent space. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 4467–4477, 2017.
- [21] Augustus Odena, Vincent Dumoulin, and Chris Olah. Deconvolution and checkerboard artifacts. *Distill*, 1(10):e3, 2016.
- [22] Chris Olah, Alexander Mordvintsev, and Ludwig Schubert. Feature visualization. *Distill*, 2(11):e7, 2017.
- [23] Omkar M Parkhi, Andrea Vedaldi, Andrew Zisserman, and CV Jawahar. Cats and dogs. In *2012 IEEE conference on computer vision and pattern recognition*, pages 3498–3505. IEEE, 2012.
- [24] Vitali Petsiuk, Abir Das, and Kate Saenko. Rise: Randomized input sampling for explanation of black-box models. *arXiv preprint arXiv:1806.07421*, 2018.
- [25] Olga Russakovsky, Jia Deng, Hao Su, Jonathan Krause, Sanjeev Satheesh, Sean Ma, Zhiheng Huang, Andrej Karpathy, Aditya Khosla, Michael Bernstein, et al. Imagenet large scale visual recognition challenge. *International journal of computer vision*, 115(3):211–252, 2015.
- [26] Ramprasaath R Selvaraju, Michael Cogswell, Abhishek Das, Ramakrishna Vedantam, Devi Parikh, and Dhruv Batra. Grad-cam: Visual explanations from deep networks via gradient-based localization. In *Proceedings of the IEEE international conference on computer vision*, pages 618–626, 2017.
- [27] Catherine Wah, Steve Branson, Peter Welinder, Pietro Perona, and Serge Belongie. The caltech-ucsd birds-200-2011 dataset. 2011.
- [28] Haofan Wang, Zifan Wang, Mengnan Du, Fan Yang, Zijian Zhang, Sirui Ding, Piotr Mardziel, and Xia Hu. Score-cam: Score-weighted visual explanations for convolutional neural networks. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition workshops*, pages 24–25, 2020.
- [29] Yuxin Wu and Kaiming He. Group normalization. In *Proceedings of the European conference on computer vision (ECCV)*, pages 3–19, 2018.
- [30] Matthew D Zeiler and Rob Fergus. Visualizing and understanding convolutional networks. In *European conference on computer vision*, pages 818–833. Springer, 2014.

## A Additional Method Details

### A.1 Detailed Diagnosis of the Two Crises

Section 2.1 of the main text identifies two algebraic pathologies of the raw VJP. We expand here on the rigorous mathematical content of each, prove the angular collapse inequality (1), and discuss when its parallel noise bound holds in practice.

#### Crisis I: zero insertion as a discrete Dirac comb.

Let a stage transition apply a stride- $s$  convolution with kernel  $W$  to a feature map  $h_{l-1}$ , producing  $h_l$ . In the forward pass this acts as decimation: only spatial positions on the stride lattice are retained. In the backward pass, the chain rule applied to  $h_l = W \circledast_{(s)} h_{l-1}$  yields

$$\nabla_{h_{l-1}} \mathcal{J} = W^T \circledast \uparrow_s (\nabla_{h_l} \mathcal{J}), \quad (16)$$

where  $\uparrow_s$  is the zero insertion upsampling operator and  $\otimes$  is standard convolution on the upsampled grid. The operator  $\uparrow_s$  produces a sparse signal whose support is the integer lattice  $s\mathbb{Z} \times s\mathbb{Z}$ ,

$$[\uparrow_s u](r, t) = \sum_{p, q} u(p, q) \delta(r - sp, t - sq). \quad (17)$$

In signal processing terms,  $\uparrow_s$  modulates the gradient continuum by a discrete two dimensional Dirac comb  $\text{III}_s$  of period  $s$ . Its Fourier dual is itself a Dirac comb of period  $1/s$ , so  $\uparrow_s u$  contains aliased high frequency replicas of  $u$  at every integer multiple of the Nyquist frequency. The transposed convolution  $W^T \otimes (\cdot)$  that follows is, generically, an interpolation kernel of finite spectral support and is mathematically incapable of removing these replicas. Consequently, every traversal of a stage boundary by the VJP injects high frequency content into the gradient signal, geometrically realised as a “nail bed” of point spikes at the stride lattice. The spatial topology that the VJP would otherwise carry is broken at exactly these boundaries.

## 428 Crisis II: rigid coupling of magnitude and phase.

429 For any pixel space VJP vector  $v \in \mathbb{R}^D$  with  $v \neq 0$ , write

$$v = \|v\| \cdot \frac{v}{\|v\|}. \quad (18)$$

430 In a magnitude phase decoupled representation, the scalar  $\|v\|$  and the unit direction  $v/\|v\|$  would be  
431 stored or normalised independently. In the raw VJP they share the same vector slot, so the spatial  
432 direction inherits no independent scaling protection. As  $v$  is propagated through deep stages, its  
433 norm  $\|v\|$  decays multiplicatively under the shrinking operator norm of the cumulative Jacobian  
434 transposes, while the additive structural noise  $\epsilon$  from Crisis I retains absolute magnitude. Once  $\|\epsilon\|$   
435 becomes comparable to  $\|v\|$ , the perturbed direction  $(v + \epsilon)/\|v + \epsilon\|$  no longer reflects the true  
436 gradient direction  $v/\|v\|$ .

## 437 Proof of the angular collapse inequality (1).

438 Decompose  $\epsilon$  along  $v$  as  $\epsilon = \epsilon_{\parallel} + \epsilon_{\perp}$  with  $\epsilon_{\parallel} = \alpha v$  for some scalar  $\alpha \in \mathbb{R}$  and  $\langle \epsilon_{\perp}, v \rangle = 0$ . Then

$$v + \epsilon = (1 + \alpha)v + \epsilon_{\perp}, \quad \|v + \epsilon\|^2 = (1 + \alpha)^2 \|v\|^2 + \|\epsilon_{\perp}\|^2. \quad (19)$$

439 Define  $\hat{v} := v/\|v\|$  and  $\hat{w} := (v + \epsilon)/\|v + \epsilon\|$ . Their inner product is

$$\langle \hat{v}, \hat{w} \rangle = \frac{(1 + \alpha)\|v\|^2}{\|v\| \sqrt{(1 + \alpha)^2 \|v\|^2 + \|\epsilon_{\perp}\|^2}} = \frac{1 + \alpha}{\sqrt{(1 + \alpha)^2 + \rho^2}}, \quad \rho := \frac{\|\epsilon_{\perp}\|}{\|v\|}. \quad (20)$$

440 Under the hypotheses  $\|\epsilon_{\parallel}\| \leq K\|v\|$  (so  $|\alpha| \leq K$  and  $1 + \alpha \leq 1 + K$ ) and  $\|\epsilon_{\perp}\| \geq \|v\|$  (so  $\rho \geq 1$ ),  
441 the right hand side is bounded above by

$$\langle \hat{v}, \hat{w} \rangle \leq \frac{1 + K}{\sqrt{(1 + K)^2 + 1}}, \quad (21)$$

442 where we used the monotonicity of  $t \mapsto t/\sqrt{t^2 + \rho^2}$  in  $t$  and the lower bound  $\rho^2 \geq 1$ . Since  $\hat{v}$  and  $\hat{w}$   
443 are unit vectors,  $\|\hat{w} - \hat{v}\|^2 = 2 - 2\langle \hat{v}, \hat{w} \rangle$ , so

$$\|\hat{w} - \hat{v}\|^2 \geq 2 - \frac{2(1 + K)}{\sqrt{(1 + K)^2 + 1}} = \frac{2(\sqrt{(1 + K)^2 + 1} - (1 + K))}{\sqrt{(1 + K)^2 + 1}}. \quad (22)$$

444 Multiplying numerator and denominator by  $\sqrt{(1 + K)^2 + 1} + (1 + K)$  yields

$$\|\hat{w} - \hat{v}\|^2 \geq \frac{2}{(\sqrt{(1 + K)^2 + 1})(\sqrt{(1 + K)^2 + 1} + (1 + K))} \geq \frac{1}{(1 + K)^2 + 1}, \quad (23)$$

445 where the second inequality uses  $\sqrt{(1 + K)^2 + 1} + (1 + K) \leq 2\sqrt{(1 + K)^2 + 1}$ . Taking the square  
446 root,

$$\|\hat{w} - \hat{v}\| \geq \frac{1}{\sqrt{(1 + K)^2 + 1}} = c > 0, \quad (24)$$

447 which is (1). The bound  $c$  is independent of  $\|v\|$ , so phase collapse persists no matter how slowly  $v$   
448 decays.

## Why the parallel noise bound holds in practice.

The hypothesis  $\|\epsilon_{\parallel}\| \leq K\|v\|$  asserts that the projection of  $\epsilon$  onto the direction of  $v$  is, in scale, no larger than  $v$  itself. We argue informally that this naturally holds for the structural noise generated by zero insertion. The signal  $v$  is the cumulative product of channel-level Jacobian transposes applied to a smooth low frequency seed (a feature map with finite spatial bandwidth). The noise  $\epsilon$  originates from the spike pattern  $\uparrow_s u$  in (16), which lives in a high frequency band concentrated near the Nyquist replicas  $|f| \geq 1/(2s)$ . Because  $W^T$  has bounded spectral mass at low frequencies, the projection of the high frequency spike pattern onto the smooth direction of  $v$  is suppressed, and empirically the projection ratio  $\|\epsilon_{\parallel}\|/\|v\|$  is bounded by a small constant  $K$  across natural images. The orthogonal component  $\epsilon_{\perp}$  carries the dominant noise energy, satisfying  $\|\epsilon_{\perp}\| \geq \|v\|$  in deep layers where  $v$  has decayed. The conditions of (1) are therefore not pathological assumptions but a generic property of the VJP cascade in modern strided CNNs.

## A.2 Existence and Geometry of the Simplex Measure

Section 2.2 defines the per-layer simplex measure  $E_{l,c}$  via  $Z_{l,c} = \text{GAP}(|h_{l,c}|)$  and  $E_{l,c} = Z_{l,c} / \sum_{c'} Z_{l,c'}$ . We note three properties.

**Existence.** Since  $|h_{l,c}| \geq 0$  pointwise, every  $Z_{l,c} \geq 0$ . The denominator  $\sum_{c'} Z_{l,c'}$  vanishes if and only if  $h_{l,c'} \equiv 0$  for every  $c'$ , that is, the encoder maps the input to the all-zero feature tensor at layer  $l$ . For any frozen encoder pretrained on natural images, this corresponds to a measure-zero set of inputs (essentially only  $X \equiv 0$  for ReLU based architectures, since at least one ReLU unit is active for any non-trivial input).  $E_{l,c}$  is therefore well-defined for every non-trivial input.

**Simplex membership.** By construction  $E_{l,c} \geq 0$  and  $\sum_c E_{l,c} = 1$  as an algebraic identity rather than an asymptotic regularisation outcome. The vector  $(E_{l,1}, \dots, E_{l,C_l})$  resides on the standard  $(C_l - 1)$ -dimensional probability simplex

$$\Delta^{C_l-1} = \left\{ p \in \mathbb{R}_{\geq 0}^{C_l} : \sum_c p_c = 1 \right\}, \quad (25)$$

and admits the interpretation of the encoder’s confidence distribution over the  $C_l$  semantic concepts at abstraction level  $l$ . It is complete and mutually exclusive: assigning more energy to one concept necessarily reduces the energy of the others.

**Polarity is removed by absolute value, not destroyed.** Modern architectures employing non-monotonic activations (GELU, Swish) or signed feature maps after BatchNorm offsets can produce  $h_{l,c}(r, s) < 0$ . Taking the absolute value before pooling guarantees  $Z_{l,c} \geq 0$  across all such architectures and treats excitatory ( $h > 0$ ) and active suppressive ( $h < 0$ ) responses equally as far as importance is concerned. The sign information itself is not discarded: it is encoded in the spatial phase  $\tilde{V}_{l,c}$  of the backward path, where it contributes to the polarity of pixel-space contrast. This is a deliberate consequence of the magnitude phase decoupling principle.

## A.3 LAC as Amplitude Stripping Operator: Geometric Details

Section 2.3 introduces the LAC and the GroupNorm strip (6). We expand on three aspects: deployment topology, the three-step geometric factorisation, and the cascade depth seen by each stage.

### Markovian Transition Guards.

The LAC modules are deployed as guards at the boundaries of the network’s Markov chain (5), not after every convolution. The justification has two layers. First, intra-stage convolutions preserve spatial resolution and operate on the same semantic manifold; their adjoints are non-strided transposed convolutions whose forward and backward operations are exact inverses up to a finite spectral tail, so the gradient flow is smooth and requires no intervention. Second, only the stride-2 stage transitions and the initial stem cliff (typically a  $7 \times 7$  stride-2 convolution followed by max pooling) actually invoke the zero insertion operator  $\uparrow_s$  during backpropagation. Placing an LAC at every convolution would introduce redundant normalisation that disturbs intra-stage smoothness while doing nothing additional to mitigate the structural shock. The exclusive deployment at boundaries is thus a minimal and topology driven design.

496 **The three geometric actions of GroupNorm.**

497 For a flattened VJP vector  $v \in \mathbb{R}^n$  with  $n = H_l W_l$ , the LAC executes the composition

$$v \mapsto v - \mu(v)\mathbf{1} \mapsto \frac{v - \mu(v)\mathbf{1}}{\sigma(v)} \mapsto \gamma_c \frac{v - \mu(v)\mathbf{1}}{\sigma(v)} + \beta_c \mathbf{1}, \quad (26)$$

498 where  $\mu(v) = (1/n)\mathbf{1}^T v$  and  $\sigma(v) = \|P_{\perp \mathbf{1}} v\|/\sqrt{n}$ . We analyse each step.

499 (a) *DC stripping*. The map  $v \mapsto v - \mu\mathbf{1} = P_{\perp \mathbf{1}} v$  is the orthogonal projection onto the zero  
500 mean hyperplane  $\mathcal{H} = \{w \in \mathbb{R}^n : \mathbf{1}^T w = 0\}$ . Geometrically, this removes the spatially-uniform  
501 component of  $v$ . The structural noise  $\epsilon$  injected by Crisis I has a non-zero average (its DC component  
502 is the sum of all spike amplitudes within a stride- $s$  tile, which is generically nonzero), so projecting  
503 onto  $\mathcal{H}$  kills the dominant scalar carrier of the noise.

504 (b) *Magnitude stripping*. The map  $w \mapsto w/\sigma(w) = w\sqrt{n}/\|w\|$  on  $\mathcal{H}$  is the radial projection onto  
505 the sphere of radius  $\sqrt{n}$  inside  $\mathcal{H}$ . This eliminates the absolute scale of the vector while preserving  
506 its direction  $w/\|w\|$  exactly. Because Crisis II rigidly couples scale and direction, this step is the  
507 algebraic act of decoupling: from this point onward, the spatial phase is stored on a manifold whose  
508 metric is independent of the encoder’s energy decay rate.

509 (c) *Learned affine*. The map  $\hat{v} \mapsto \gamma_c \hat{v} + \beta_c \mathbf{1}$  is a one dimensional rescaling on the sphere followed  
510 by a uniform shift. The scale  $\gamma_c$  controls only the global brightness contribution of channel  $c$ ; the  
511 shift  $\beta_c$  is a global pixel offset. Crucially, since  $\gamma_c$  acts isotropically on the unit sphere and  $\beta_c$  acts  
512 only on the constant subspace, neither parameter has any spatial selectivity, a property formalised in  
513 Proposition B.8.

514 The dominant noise carriers of Crisis I are the systematic DC drift (annihilated by step (a)) and the  
515 absolute magnitude scale (annihilated by step (b)). Step (c) cannot undo these annihilations because  
516 its parameters are scalar. The noise is therefore not smoothed away by post hoc filtering; it is removed  
517 by an exact algebraic projection onto a noise-free subspace.

518 **Cascade depth seen by each source stage.**

519 Suppose the encoder has  $L$  feature stages plus a stem. A feature at stage  $l$  traverses, during the  
520 backward cascade (7), exactly the LAC modules

$$\Psi_{\text{LAC}_l}, \Psi_{\text{LAC}_{l-1}}, \dots, \Psi_{\text{LAC}_0}, \Psi_{\text{LAC}_{\text{stem}}}, \quad (27)$$

521 which is  $l + 2$  LAC modules in total. The deepest stage  $L - 1$  exercises every LAC in the system;  
522 any shallower stage exercises a strict subset. This nested structure is what enables the “train once,  
523 infer at every depth” property of the framework, and is what underlies the path invariance result of  
524 Proposition B.8.

525 **A.4 Training Strategy and Inference Spectrum: Detailed Analysis**

526 Section 2.4 adopts the single path objective  $\mathcal{L} = \|X - \hat{X}_{L-1}\|_1$ . We motivate the two design choices  
527 that this objective embodies: training on the deepest stage only, and using the  $L_1$  rather than  $L_2$  norm.  
528 We also describe the inference time semantic inversion spectrum.

529 **Why train on the deepest stage only.**

530 An apparent alternative is the multi-path summation

$$\mathcal{L}_{\text{multi}} = \sum_{l=0}^{L-1} \|X - \hat{X}_l\|_1. \quad (28)$$

531 Although natural at first sight, this objective induces pathological gradient competition for the  
532 shared lower level LACs ( $\Psi_{\text{LAC}_0}$  and  $\Psi_{\text{LAC}_{\text{stem}}}$ ). For shallow stages, the spatial signal in  $\hat{X}_l$  is  
533 already richly structured because the input has been compressed through fewer downsampling cliffs;  
534 the corresponding reconstruction task is structurally easier. The shared LAC parameter gradient  
535  $\partial \mathcal{L}_l / \partial \theta$  from a shallow stage is therefore both larger in expected magnitude and biased toward a  
536 less demanding spatial decompression regime. Under (28), the shared LAC parameters would be

systematically driven away from the configuration required for the most demanding deepest-stage reconstruction.

Training on  $\mathcal{L} = \|X - \hat{X}_{L-1}\|_1$  avoids this competition entirely. By the cascade composition (7), the adjoint path of stage  $L - 1$  traverses every LAC in the system, providing complete gradient coverage of all trainable parameters with no LAC left in a gradient dead zone. The shared lower level LACs are thus trained under the most stringent regime; their performance on the easier shallow stages is, by virtue of Proposition B.8, structurally guaranteed to be no worse.

#### **Inference time semantic inversion spectrum.**

Although training touches only the deepest path, the trained LAC immediately supports inference at every depth. For any  $l < L - 1$ , the cascade required to compute  $\hat{X}_l$  uses the LAC modules  $\Psi_{\text{LAC}_l}, \dots, \Psi_{\text{LAC}_0}, \Psi_{\text{LAC}_{\text{stem}}}$ , all of which are a strict subset of the deepest path’s cascade and have therefore been fully optimised. Reading off the family  $\{\hat{X}_l\}_{l=0}^{L-1}$  at inference produces what we call the Semantic Inversion Spectrum, a depth indexed sequence of pixel space images that interpolates from high frequency structural textures at shallow layers, where the encoder has not yet abstracted away the local pixel statistics, to spatially sparse categorical activation loci at deep layers, where only object discriminative features survive. The spectrum directly answers the question “what is the visual structure of this image when interpreted strictly through the isolated semantics of layer  $l$ ?”, and it does so without any retraining.

### **A.5 Shared Adjoint Bridges and Endogenous Consistency**

The cascade (7) shares its lower level modules ( $\Psi_{\text{LAC}_0}$  and  $\Psi_{\text{LAC}_{\text{stem}}}$ ) across all source stages  $l \geq 0$ . We discuss why this sharing is structurally consistent rather than a source of cross layer interference, and how it underpins the framework’s plug and play extension to arbitrary downstream heads.

#### **Why consistency holds across source stages.**

The shared modules are trained, by the single path objective of Appendix A.4, under the most demanding deep layer reconstruction regime. When the same modules are subsequently used to process gradient signals arriving from a shallower source stage  $l' < L - 1$ , two structural facts apply.

First, the per-channel GroupNorm with `num_groups = num_channels` exactly eliminates the dominant source of path dependent variation, namely per-channel spatial mean and variance, before the affine parameters act. Specifically, for any two source stages  $A$  and  $B$ , the standardised inputs  $\hat{v}_{b,c}^{(A)}$  and  $\hat{v}_{b,c}^{(B)}$  at any shared LAC level  $b$  both have spatial mean 0 and spatial second moment 1 as exact algebraic identities that do not involve any trainable parameter. This is the first-and-second-moment invariance formalised in Proposition B.8(a).

Second, the scalar affine parameters  $(\gamma_c^{(b)}, \beta_c^{(b)})$  have exactly one degree of freedom per channel and are therefore structurally incapable of differentially responding to residual higher-order spatial discrepancies between deep path and shallow path signals. The map induced by  $\gamma_c^{(b)}$  is an isotropic scaling on the zero mean hyperplane (it dilates every spatial direction by the same factor); the map induced by  $\beta_c^{(b)}$  acts only on the constant subspace. Neither has spatial selectivity, a property formalised as the scalar insensitivity in Proposition B.8(b).

Together, these two properties imply that the shared LAC modules treat deep path and shallow path signals on equal footing up to their first two moments; the only differences allowed by the parametric capacity are channel level isotropic rescaling, which carries no spatial structural distinction.

#### **Quality monotonicity in source depth.**

Because the shared modules were optimised under the deepest stage regime, where the spatial reconstruction task is hardest, their behaviour on shallower stage signals (which already carry richer spatial information) is structurally guaranteed to be no worse. Concretely, any reconstruction error  $\hat{X}_l$  for  $l < L - 1$  inherits the error properties of the deepest path with respect to the shared lower level LACs, while gaining additional spatial fidelity from the richer signal at stage  $l$ . The framework therefore natively supports plug and play visualisation for any layer and any channel

without retraining; cross layer explanatory consistency is an endogenous property of the shared topology, not an artefact of post hoc calibration.

#### Extension to arbitrary downstream heads.

The shared topology property also extends the framework, without retraining, to arbitrary differentiable downstream heads beyond the linear classifier. Concretely, given any differentiable module  $\mathcal{D}$  acting on the deepest feature map  $h_{L-1}$  and any scalar readout  $Q$  of  $\mathcal{D}$ , the pixel space visualisation of  $Q$  is obtained by computing the single step VJP  $\partial Q / \partial h_{L-1}$  and feeding it as the seed into the trained LAC cascade. The moment and direction guarantees of Proposition B.8 ensure the resulting visualisation is no less faithful than for a linear classifier. We exploit this property in Section 3.5, where three layers of self-attention are stacked on top of the frozen encoder and per layer / per token visualisations are produced through the same trained LAC cascade with no parameter modification.

## B Theoretical Statements and Proofs

This appendix collects the formal statements and proofs of all theoretical results that underpin the framework. Sections B.1–B.3 present three foundational results (Lemma B.1, Proposition B.2, Proposition B.3); Sections B.4–B.7 prove the two main-text guarantees (Theorem 1 and Corollary 1) along with two intermediate results (Theorem B.5, Corollary B.6); Section B.8 treats the path invariance property of shared LAC bridges (Proposition B.8) which underpins the framework’s plug-and-play extension to arbitrary downstream heads.

### B.1 Lemma B.1: Training Dynamics of Affine Parameters

**Lemma 1** (B.1: Training Dynamics of Affine Parameters). *Consider a LAC decoder trained with the  $L_1$  reconstruction loss on a diverse image distribution satisfying  $\mathbb{E}[X] \approx \mathbf{0}$  and approximate spatial symmetry about zero. Let  $(\gamma_c^*, \beta_c^*)$  denote the learned GroupNorm affine parameters at convergence. Then:*

- (i)  $\beta_c^* \rightarrow 0$  for every channel  $c$ ;
- (ii)  $\gamma_c^* > 0$  for every active channel  $c \in \mathcal{C}^+ := \{c : E_{l,c} > 0\}$ .

*Proof.* We establish the two parts separately.

**Part (i):**  $\beta_c^* \rightarrow 0$ . By Proposition 2 (Eq. (43)), the decoded output at layer  $l$  admits the decomposition

$$\hat{X}_l = \underbrace{\sum_{c \in \mathcal{C}^+} E_{l,c} \gamma_c \sqrt{H_l W_l} \frac{P_{\perp \mathbf{1}} v_c}{\|P_{\perp \mathbf{1}} v_c\|_2}}_{S_l \text{ (spatially varying)}} + \underbrace{\left( \sum_{c=1}^{C_l} E_{l,c} \beta_c \right) \mathbf{1}}_{b_l \mathbf{1} \text{ (global constant offset)}}, \quad (29)$$

where  $v_c = \text{vec}(h_{l,c})$  and  $b_l = \sum_c E_{l,c} \beta_c$ .

The spatially varying component  $S_l$  is orthogonal to  $\mathbf{1}$  by construction (each summand lies in the range of  $P_{\perp \mathbf{1}}$ ), so  $S_l$  and  $b_l \mathbf{1}$  reside in complementary orthogonal subspaces. Consequently, the  $L_1$  reconstruction loss decomposes as

$$\mathcal{L} = \mathbb{E}_X [\|X - \hat{X}_l\|_1] \geq \mathbb{E}_X [\|P_1(X - \hat{X}_l)\|_1] = \mathbb{E}_X [\|\bar{X} \mathbf{1} - b_l \mathbf{1}\|_1], \quad (30)$$

where  $\bar{X} = \frac{1}{n} \mathbf{1}^T X$  and  $P_1 = \frac{1}{n} \mathbf{1} \mathbf{1}^T$  is the projection onto the constant subspace. Minimising the right-hand side over the scalar  $b_l$  yields the classical  $L_1$  location problem: the minimiser is the conditional median of  $\bar{X}$  over the training distribution.

Under the assumption that the training distribution is diverse with  $\mathbb{E}[X] \approx \mathbf{0}$  and approximately symmetric about zero, the pixelwise median of the target images is also approximately zero, giving  $b_l^* \approx 0$ .

To argue that  $\beta_c^* \rightarrow 0$  individually rather than merely their weighted sum: during gradient-based training each  $\beta_c$  receives the gradient signal

$$\frac{\partial \mathcal{L}}{\partial \beta_c} = E_{l,c} \frac{\partial \mathcal{L}}{\partial b_l}, \quad (31)$$

since  $b_l$  depends linearly on each  $\beta_c$  with coefficient  $E_{l,c}$ . For active channels ( $E_{l,c} > 0$ ), the gradient  $\partial \mathcal{L} / \partial \beta_c$  shares the sign of  $\partial \mathcal{L} / \partial b_l$  and drives  $\beta_c$  toward zero at a rate proportional to  $E_{l,c}$ . For inactive channels ( $E_{l,c} = 0$ ),  $\beta_c$  receives no gradient and remains at its initialisation value zero (PyTorch default). In either case  $\beta_c^* \approx 0$  at convergence.

**Part (ii):  $\gamma_c^* > 0$ .** We argue by examining the three regimes of  $\gamma_c$  and showing that only  $\gamma_c > 0$  is consistent with loss minimisation.

*Case  $\gamma_c = 0$ .* The affine output reduces to  $\text{GN}(v_c) = \beta_c \mathbf{1}$  by Eq. (43), which is spatially uniform. The contribution of channel  $c$  to  $\hat{X}_l$  then carries no spatial structure whatsoever, forcing a strictly positive reconstruction error  $\varepsilon_0 > 0$  that no other parameter associated with channel  $c$  can reduce. Hence  $\gamma_c = 0$  is sub-optimal.

*Case  $\gamma_c < 0$ .* By Property (P3) of Proposition 2, the spatial direction of the GroupNorm output (ignoring bias) is  $P_{\perp \mathbf{1}} v_c / \|P_{\perp \mathbf{1}} v_c\|_2$  when  $\gamma_c > 0$ , and its negation when  $\gamma_c < 0$ . In the LAC decoder,  $v_c$  encodes information derived from the activation-energy gradient  $\nabla_X \Phi_{l,c}(X)$  via Proposition 1, so the “correct” spatial phase for reconstructing  $X$  is the one aligned with this gradient. Negating the phase systematically reverses all spatial contrast carried by channel  $c$  (bright regions become dark and vice versa), strictly increasing the  $L_1$  error relative to the  $\gamma_c > 0$  case for every input with non-trivial activation in channel  $c$ . Formally, letting  $\hat{X}_l(\gamma_c)$  denote the reconstruction parameterised by  $\gamma_c$  with all other parameters fixed,

$$\|X - \hat{X}_l(-|\gamma_c|)\|_1 > \|X - \hat{X}_l(+|\gamma_c|)\|_1 \quad (32)$$

whenever channel  $c$ ’s contribution has non-zero projection onto the residual  $X - \hat{X}_l^{c^c}$ , which holds generically for active channels.

*Case  $\gamma_c > 0$ .* The spatial direction of the GroupNorm output is aligned with the zero-mean activation-energy gradient, providing the maximum possible reduction in  $L_1$  error per unit of  $|\gamma_c|$ . The loss is monotonically decreasing in  $\gamma_c$  for small positive values and the global minimiser must lie in the interior of the positive half-line.

Combining the three cases, every active channel satisfies  $\gamma_c^* > 0$  at convergence.  $\square$

## B.2 Proposition B.2: VJP as Activation Energy Gradient

**Proposition 1** (B.2: VJP as Activation Energy Gradient). *Let  $h_l(X) \in \mathbb{R}^{C_l \times H_l \times W_l}$  denote the encoder feature tensor at layer  $l$ , viewed as a differentiable function of the input  $X \in \mathbb{R}^{3 \times H \times W}$ . Define the channel-wise activation energy*

$$\Phi_{l,c}(X) = \frac{1}{2} \|h_{l,c}(X)\|^2 = \frac{1}{2} \sum_{r=1}^{H_l} \sum_{s=1}^{W_l} h_{l,c}(r, s; X)^2, \quad (33)$$

and the channel-selective VJP

$$\text{VJP}_{l,c} = \left( \frac{\partial h_l}{\partial X} \right)^T (e_c \odot h_l), \quad (34)$$

where  $e_c \in \mathbb{R}^{C_l}$  is the one-hot indicator selecting channel  $c$  and  $\odot$  denotes channel-wise broadcasting. Then

$$\text{VJP}_{l,c} = \nabla_X \Phi_{l,c}(X). \quad (35)$$

*Proof. Step 1 (Reduction to a single-channel VJP).* Decompose the full Jacobian  $J_l = \partial h_l / \partial X \in \mathbb{R}^{(C_l H_l W_l) \times (3HW)}$  into channel blocks  $J_l = [J_{l,1}^T; \dots; J_{l,C_l}^T]^T$ , where  $J_{l,c} = \partial h_{l,c} / \partial X \in \mathbb{R}^{(H_l W_l) \times (3HW)}$  is the Jacobian of the  $c$ -th channel map alone. Because the seed  $e_c \odot h_l$  is supported exclusively on channel  $c$ ,

$$\text{VJP}_{l,c} = J_l^T (e_c \odot h_l) = \sum_{c'=1}^{C_l} J_{l,c'}^T [e_c \odot h_l]_{c'} = J_{l,c}^T h_{l,c}, \quad (36)$$

where all terms with  $c' \neq c$  vanish.

661 **Step 2 (Gradient of the energy functional).** Differentiating (33) with respect to pixel  $(i, j)$ :

$$\frac{\partial \Phi_{l,c}}{\partial X(i, j)} = \sum_{r=1}^{H_l} \sum_{s=1}^{W_l} h_{l,c}(r, s) \frac{\partial h_{l,c}(r, s)}{\partial X(i, j)}. \quad (37)$$

662 The exchange of differentiation and finite summation is exact. Collecting (37) over all pixels,

$$\nabla_X \Phi_{l,c}(X) = J_{l,c}^T h_{l,c}. \quad (38)$$

663 **Step 3 (Identification).** Comparing (36) and (38), both right-hand sides equal  $J_{l,c}^T h_{l,c}$ , hence  
 664  $\text{VJP}_{l,c} = \nabla_X \Phi_{l,c}(X)$ .  $\square$

665 **Remark 1** (Scope of validity). *The identity (35) holds for any encoder whose forward map*  
 666  *$X \mapsto h_l(X)$  is Fréchet differentiable at  $X$ . In standard convolutional encoders composed of*  
 667 *affine convolutions, batch/group normalisation, and piecewise linear activations, differentiability*  
 668 *holds outside the measure-zero set of ReLU kink boundaries, and the VJP computed by automatic*  
 669 *differentiation coincides with (35) wherever defined.*

### 670 B.3 Proposition B.3: Algebraic Structure of GroupNorm

671 **Proposition 2** (B.3: Algebraic Structure of GroupNorm). *Let  $h_{l,c}(X) \in \mathbb{R}^{H_l \times W_l}$  be the spatial*  
 672 *activation map of channel  $c$  at layer  $l$ , and let  $v = \text{vec}(h_{l,c}) \in \mathbb{R}^n$  with  $n = H_l W_l$  denote its*  
 673 *spatial flattening. Suppose GroupNorm is applied with  $\text{num\_groups} = C_l$  (each channel forms its*  
 674 *own group) and learnable affine parameters  $(\gamma_c, \beta_c) \in \mathbb{R} \times \mathbb{R}$ . Define the orthogonal de-meaning*  
 675 *projector*

$$P_{\perp \mathbf{1}} = I_n - \frac{1}{n} \mathbf{1} \mathbf{1}^T. \quad (39)$$

676 *Then, for every non-constant  $v$  (i.e.  $P_{\perp \mathbf{1}} v \neq \mathbf{0}$ ), the GroupNorm output satisfies:*

677 (P1) **Fixed L2 magnitude.**  $\|\gamma_c \hat{v}\|_2 = |\gamma_c| \sqrt{H_l W_l}$ , independent of  $v$ .

678 (P2) **DC component.**  $\frac{1}{n} \mathbf{1}^T \text{GN}(v) = \beta_c$ .

679 (P3) **Spatial direction preservation.** *The zero-mean component of  $\text{GN}(v)$  points along*  
 680  $P_{\perp \mathbf{1}} v / \|P_{\perp \mathbf{1}} v\|_2$ .

681 *Proof.* Throughout we write  $n = H_l W_l$  and suppress subscripts.

682 **Step 1 (Group structure).** With  $\text{num\_groups} = C_l$ , each normalisation group consists of a single  
 683 channel, so the statistics are computed over the  $n$  spatial positions of that channel alone.

684 **Step 2 (Mean and variance in projector form).** The channel mean and variance are

$$\mu = \frac{1}{n} \mathbf{1}^T v, \quad (40)$$

685

$$\sigma^2 = \frac{1}{n} \sum_{k=1}^n (v_k - \mu)^2 = \frac{1}{n} \|v - \mu \mathbf{1}\|_2^2 = \frac{1}{n} \|P_{\perp \mathbf{1}} v\|_2^2, \quad (41)$$

686 since  $v - \mu \mathbf{1} = P_{\perp \mathbf{1}} v$ . Taking the positive square root,  $\sigma = \|P_{\perp \mathbf{1}} v\|_2 / \sqrt{n}$ .

687 **Step 3 (Normalised output).** The centred-and-scaled output before the affine transform is

$$\hat{v} = \frac{v - \mu \mathbf{1}}{\sigma} = \frac{P_{\perp \mathbf{1}} v}{\|P_{\perp \mathbf{1}} v\|_2 / \sqrt{n}} = \sqrt{n} \frac{P_{\perp \mathbf{1}} v}{\|P_{\perp \mathbf{1}} v\|_2}. \quad (42)$$

688 **Step 4 (Full affine output).** Applying the learnable affine,

$$\text{GN}(v) = \gamma_c \hat{v} + \beta_c \mathbf{1} = \gamma_c \sqrt{H_l W_l} \frac{P_{\perp \mathbf{1}} v}{\|P_{\perp \mathbf{1}} v\|_2} + \beta_c \mathbf{1}. \quad (43)$$

689 **Step 5 (Verification).**

690 (P1): From (42), the vector  $P_{\perp \mathbf{1}} v / \|P_{\perp \mathbf{1}} v\|_2$  is unit norm, so  $\|\gamma_c \hat{v}\|_2 = |\gamma_c| \sqrt{H_l W_l}$ , independent  
 691 of  $v$ .

(P2): Since  $P_{\perp 1}$  projects onto  $\mathbf{1}^\perp$ ,  $\mathbf{1}^T P_{\perp 1} v = 0$  for every  $v$ , hence

$$\frac{1}{n} \mathbf{1}^T \text{GN}(v) = \frac{\gamma_c \sqrt{n}}{n} \cdot 0 + \frac{\beta_c}{n} \mathbf{1}^T \mathbf{1} = \beta_c. \quad (44)$$

(P3): The zero-mean component of  $\text{GN}(v)$  is

$$P_{\perp 1} \text{GN}(v) = \gamma_c \sqrt{H_l W_l} \frac{P_{\perp 1} v}{\|P_{\perp 1} v\|_2} + \beta_c \underbrace{P_{\perp 1} \mathbf{1}}_{=0} = \gamma_c \sqrt{H_l W_l} \frac{P_{\perp 1} v}{\|P_{\perp 1} v\|_2}, \quad (45)$$

using the idempotence  $P_{\perp 1}^2 = P_{\perp 1}$  and  $P_{\perp 1} \mathbf{1} = \mathbf{0}$ . For  $\gamma_c > 0$ , the unit direction is precisely  $P_{\perp 1} v / \|P_{\perp 1} v\|_2$ .  $\square$

#### B.4 Proof of Theorem 1 (Spatial Fidelity)

We prove Theorem 1 from the main text:  $\text{supp}(\text{VJP}_{l,c}) \subseteq \text{EF}_{l,c}(X)$ .

*Proof. Step 1: Expansion of the VJP.* By Proposition 1,  $\Phi_{l,c}(X) = \frac{1}{2} \|h_{l,c}\|_F^2$ , and

$$\text{VJP}_{l,c}(i, j) = [\nabla_X \Phi_{l,c}(X)]_{i,j} = \sum_{r=1}^{H_l} \sum_{s=1}^{W_l} h_{l,c}(r, s) \frac{\partial h_{l,c}(r, s)}{\partial X(i, j)}. \quad (46)$$

Each summand is the product of  $h_{l,c}(r, s) \geq 0$  and the Jacobian entry  $\partial h_{l,c}(r, s) / \partial X(i, j)$ .

**Step 2: Analysis for pixels  $(i, j) \notin \text{EF}_{l,c}(X)$ .** Fix  $(i, j) \notin \text{EF}_{l,c}(X)$ . By definition, for every spatial position  $(r, s)$  at which  $h_{l,c}(r, s) > 0$ , the pixel  $(i, j)$  does not causally influence that activation. We show each summand in (46) vanishes by considering two cases.

*Case A:  $h_{l,c}(r, s) > 0$  (active position).* Since  $(i, j) \notin \text{EF}_{l,c}(X)$ , the pixel  $X(i, j)$  does not lie in the receptive field of any active unit at  $(r, s)$ . In a ReLU network, the local input-output mapping at an active ReLU is the identity, so  $\partial h_{l,c}(r, s) / \partial X(i, j)$  reduces to the corresponding entry of the composed linear map through the active path. Because  $(i, j)$  is outside the receptive field of  $(r, s)$ , this linear map has a structural zero at the  $(i, j)$ -entry:

$$\frac{\partial h_{l,c}(r, s)}{\partial X(i, j)} = 0, \quad (47)$$

so the summand vanishes:  $h_{l,c}(r, s) \cdot 0 = 0$ .

*Case B:  $h_{l,c}(r, s) = 0$  (inactive position).* We further distinguish two sub-cases on the pre-activation  $z_{l,c}(r, s)$ , the input to the final ReLU producing  $h_{l,c}(r, s) = \text{ReLU}(z_{l,c}(r, s))$ .

*Sub-case B1:  $z_{l,c}(r, s) < 0$ .* The ReLU output is locally constant at zero, so  $\partial h_{l,c}(r, s) / \partial z_{l,c}(r, s) = 0$ . By the chain rule,  $\partial h_{l,c}(r, s) / \partial X(i, j) = 0$ . The summand is  $0 \cdot 0 = 0$ .

*Sub-case B2:  $z_{l,c}(r, s) = 0$ .* The ReLU is non-differentiable at zero. Following the standard convention adopted by all major automatic-differentiation frameworks (PyTorch, JAX, TensorFlow), the sub-gradient at  $z = 0$  is set to zero, again yielding  $\partial h_{l,c}(r, s) / \partial X(i, j) = 0$  via the chain rule, and  $h_{l,c}(r, s) \cdot 0 = 0$ .

In both sub-cases the product  $h_{l,c}(r, s) \cdot \partial h_{l,c}(r, s) / \partial X(i, j)$  vanishes.

**Step 3: Conclusion.** Combining Cases A and B, every summand in (46) is zero whenever  $(i, j) \notin \text{EF}_{l,c}(X)$ , so  $\text{VJP}_{l,c}(i, j) = 0$  for all such  $(i, j)$  and  $\text{supp}(\text{VJP}_{l,c}) \subseteq \text{EF}_{l,c}(X)$ .  $\square$

**Remark 2** (Tightness of the inclusion). *The inclusion is generally strict: a pixel  $(i, j) \in \text{EF}_{l,c}(X)$  may still yield  $\text{VJP}_{l,c}(i, j) = 0$  due to cancellations across multiple active positions or specific weight configurations. Such exact cancellations are measure-zero events in the space of weights and inputs, so equality  $\text{supp}(\text{VJP}_{l,c}) = \text{EF}_{l,c}(X)$  holds generically.*

**Remark 3** (Extension beyond ReLU). *The proof relies on (a) outputs are non-negative, ensuring  $h_{l,c}(r, s) \geq 0$ ; (b) the derivative is zero whenever the output is zero. Any activation satisfying both (e.g. Leaky-ReLU with  $\alpha \rightarrow 0$ ) admits the same guarantee. For activations violating (a), the support containment still holds approximately when the negative activations are small in magnitude, which is empirically the case for encoders pretrained on natural images.*

## 729 B.5 Theorem B.5: Feature Visualization Equivalence

730 **Theorem 3** (B.5: Feature Visualization Equivalence, Approximate). *Let  $\hat{X}_l = \sum_{c=1}^{C_l} E_{l,c} \tilde{V}_{l,c}$  denote*  
 731 *the layer- $l$  output of the LAC decoder, where  $\tilde{V}_{l,c}$  is the full LAC cascade output (Section 2.3). For*  
 732 *each active channel  $c \in \mathcal{C}^+ := \{c : E_{l,c} > 0\}$ , define the ideal feature visualisation direction*

$$\text{FV}_{l,c}(X) := \frac{P_{\perp \mathbf{1}} \nabla_X \Phi_{l,c}(X)}{\|P_{\perp \mathbf{1}} \nabla_X \Phi_{l,c}(X)\|_2}, \quad (48)$$

733 *the effective semantic weight  $\tilde{E}_{l,c} := E_{l,c} \gamma_c^* \sqrt{n}$  with  $n = H_0 W_0$ , and the effective per-channel*  
 734 *direction error*

$$\epsilon_{l,c}^{\text{DC}} := \gamma_c^* \sqrt{n} \left\| \frac{P_{\perp \mathbf{1}} f_{l,c}}{\|P_{\perp \mathbf{1}} f_{l,c}\|_2} - \text{FV}_{l,c}(X) \right\|_2, \quad (49)$$

735 *where  $f_{l,c}$  is the signal entering the final GroupNorm of the LAC cascade (defined in (52)). Under*  
 736 *the convergence conditions of Lemma 1 and the algebraic structure of Proposition 2, the synthesised*  
 737 *image admits the decomposition*

$$\hat{X}_l = \sum_{c \in \mathcal{C}^+} \tilde{E}_{l,c} \text{FV}_{l,c}(X) + R_l, \quad (50)$$

738 *with the residual bounded in  $\ell_\infty$  by*

$$\|R_l\|_\infty \leq \sum_{c=1}^{C_l} E_{l,c} |\beta_c^*| + \sum_{c=1}^{C_l} E_{l,c} \epsilon_{l,c}^{\text{DC}}. \quad (51)$$

739 *Proof.* The argument proceeds in five steps.

740 **Step 1 (Algebraic unrolling of the LAC cascade).** The cascade terminates with  $\Psi_{\text{LAC}_{\text{stem}}}$ , which  
 741 applies per-channel GroupNorm followed by affine  $(\gamma_c^*, \beta_c^*)$ . Let  $f_{l,c} \in \mathbb{R}^n$  ( $n = H_0 W_0$ ) denote the  
 742 signal entering this final GroupNorm:

$$f_{l,c} := [J_{\text{stem}}^T \circ \Psi_{\text{LAC}_0} \circ J_0^T \circ \dots \circ \Psi_{\text{LAC}_l} \circ J_l^T](\text{seed}_{l,c}). \quad (52)$$

743 Unrolling the recursion: let  $u_k$  denote the gradient signal entering the intermediate LAC at stage  $k$ .  
 744 By Proposition 2, the intermediate LAC acts as  $\Psi_{\text{LAC}_k}(u_k) = c_k(u_k - \mu_k \mathbf{1}) + \beta_k \mathbf{1}$ , with  $c_k > 0$   
 745 and  $\mu_k = \frac{1}{n_k} \mathbf{1}^T u_k$ . Substituting into the recursive pullback  $u_{k-1} = J_{k-1}^T \Psi_{\text{LAC}_k}(u_k)$ :

$$u_{k-1} = c_k J_{k-1}^T u_k + (\beta_k - c_k \mu_k) J_{k-1}^T \mathbf{1}. \quad (53)$$

746 Each intermediate LAC scales the backward signal by  $c_k$  (direction preserving) but injects an additive  
 747 perturbation proportional to  $J_{k-1}^T \mathbf{1}$ . Because  $J_{k-1}^T$  embeds sparse ReLU masks,  $J_{k-1}^T \mathbf{1}$  is not a  
 748 constant vector, causing spatial mean-to-variation leakage. Unrolling to the input resolution:

$$f_{l,c} = \left( \prod_k c_k \right) \nabla_X \Phi_{l,c}(X) + M_{l,c}, \quad (54)$$

749 where  $\nabla_X \Phi_{l,c}(X)$  is the exact single-step gradient (Proposition 1) and  $M_{l,c}$  is the cumulative  
 750 structural noise from the intermediate mean-leakage terms.

751 For any active channel  $c \in \mathcal{C}^+$ ,  $E_{l,c} > 0$  implies  $h_{l,c}(X) \neq 0$ , so under generic conditions  
 752  $P_{\perp \mathbf{1}} f_{l,c} \neq \mathbf{0}$ . Applying Proposition 2 to the final GroupNorm:

$$\tilde{V}_{l,c} = \gamma_c^* \sqrt{n} \frac{P_{\perp \mathbf{1}} f_{l,c}}{\|P_{\perp \mathbf{1}} f_{l,c}\|_2} + \beta_c^* \mathbf{1}. \quad (55)$$

753 **Step 2 (Direction decomposition).** Define the cascade direction and ideal direction:

$$d_{l,c}^{\text{cas}} := \frac{P_{\perp \mathbf{1}} f_{l,c}}{\|P_{\perp \mathbf{1}} f_{l,c}\|_2}, \quad \text{FV}_{l,c}(X) := \frac{P_{\perp \mathbf{1}} \nabla_X \Phi_{l,c}(X)}{\|P_{\perp \mathbf{1}} \nabla_X \Phi_{l,c}(X)\|_2}. \quad (56)$$

754 Applying  $P_{\perp \mathbf{1}}$  to (54) shows  $P_{\perp \mathbf{1}} f_{l,c}$  and  $P_{\perp \mathbf{1}} \nabla_X \Phi_{l,c}(X)$  are misaligned strictly because of  
 755  $P_{\perp \mathbf{1}} M_{l,c}$ .

756 Define the direction deviation

$$\eta_{l,c} := d_{l,c}^{\text{cas}} - \text{FV}_{l,c}(X), \quad \epsilon_{l,c}^{\text{DC}} := \gamma_c^* \sqrt{n} \|\eta_{l,c}\|_2. \quad (57)$$

757 Since  $d_{l,c}^{\text{cas}}$  and  $\text{FV}_{l,c}$  are unit vectors,  $\|\eta_{l,c}\|_2 \in [0, 2]$ , so  $\epsilon_{l,c}^{\text{DC}} \in [0, 2\gamma_c^* \sqrt{n}]$ .

758 Substituting  $d_{l,c}^{\text{cas}} = \text{FV}_{l,c}(X) + \eta_{l,c}$  into (55):

$$\tilde{V}_{l,c} = \gamma_c^* \sqrt{n} [\text{FV}_{l,c}(X) + \eta_{l,c}] + \beta_c^* \mathbf{1}. \quad (58)$$

759 **Step 3 (Channel superposition).** Substituting (58) into  $\hat{X}_l = \sum_c E_{l,c} \tilde{V}_{l,c}$  and using  $\tilde{E}_{l,c} =$   
760  $E_{l,c} \gamma_c^* \sqrt{n}$ :

$$\hat{X}_l = \underbrace{\sum_{c=1}^{C_l} \tilde{E}_{l,c} \text{FV}_{l,c}(X)}_{\text{principal}} + \underbrace{\sum_{c=1}^{C_l} \tilde{E}_{l,c} \eta_{l,c}}_{R_l^{\text{DC}}} + \underbrace{\left( \sum_{c=1}^{C_l} E_{l,c} \beta_c^* \right) \mathbf{1}}_{R_l^\beta}. \quad (59)$$

761 **Step 4 (Restriction to active channels).** Channels with  $E_{l,c} = 0$  contribute zero to all three sums.

762 By Lemma 1(ii),  $\gamma_c^* > 0$  for every active channel, so  $\tilde{E}_{l,c} > 0 \iff E_{l,c} > 0$ . Restricting the  
763 principal sum to  $\mathcal{C}^+$ :

$$\hat{X}_l = \sum_{c \in \mathcal{C}^+} \tilde{E}_{l,c} \text{FV}_{l,c}(X) + R_l, \quad R_l := R_l^{\text{DC}} + R_l^\beta. \quad (60)$$

764 **Step 5 (Residual bound in  $\ell_\infty$ ).**

765 *Affine bias residual.*  $R_l^\beta = \left( \sum_c E_{l,c} \beta_c^* \right) \mathbf{1}$  is spatially uniform, so

$$\|R_l^\beta\|_\infty = \left| \sum_{c=1}^{C_l} E_{l,c} \beta_c^* \right| \leq \sum_{c=1}^{C_l} E_{l,c} |\beta_c^*|. \quad (61)$$

766 *Cascade direction residual.* By the  $\ell_\infty$  triangle inequality,

$$\|R_l^{\text{DC}}\|_\infty \leq \sum_{c=1}^{C_l} \tilde{E}_{l,c} \|\eta_{l,c}\|_\infty \leq \sum_{c=1}^{C_l} \tilde{E}_{l,c} \|\eta_{l,c}\|_2. \quad (62)$$

767 Since  $\tilde{E}_{l,c} \|\eta_{l,c}\|_2 = E_{l,c} (\gamma_c^* \sqrt{n} \|\eta_{l,c}\|_2) = E_{l,c} \epsilon_{l,c}^{\text{DC}}$  for active channels (and both sides vanish for  
768 inactive channels),

$$\|R_l^{\text{DC}}\|_\infty \leq \sum_{c=1}^{C_l} E_{l,c} \epsilon_{l,c}^{\text{DC}}. \quad (63)$$

769 *Combined.* Summing (61) and (63) yields

$$\|R_l\|_\infty \leq \sum_{c=1}^{C_l} E_{l,c} |\beta_c^*| + \sum_{c=1}^{C_l} E_{l,c} \epsilon_{l,c}^{\text{DC}}. \quad (64)$$

770 By Lemma 1(i),  $|\beta_c^*| \rightarrow 0$  for all channels.  $\square$

771 **Remark 4** (Two residual components). *The residual  $R_l$  decomposes into two qualitatively distinct*  
772 *contributions.  $R_l^\beta$  is a spatially constant offset that affects only global brightness (its spatial*  
773 *gradient is zero) and vanishes as  $\beta_c^* \rightarrow 0$ .  $R_l^{\text{DC}}$  is a spatially structured perturbation arising*  
774 *from the intermediate LAC modules altering the backward signal through mean removal, variance*  
775 *standardisation, and affine rescaling at each manifold boundary; this residual persists even under*  
776 *perfect bias convergence ( $\beta_c^* = 0$  for all  $c$ ) and is intrinsic to the multi-stage cascade architecture.*

777 **Remark 5** (Why  $\epsilon_{l,c}^{\text{DC}}$  is small in practice). *Although the trivial bound is  $\epsilon_{l,c}^{\text{DC}} \leq 2\gamma_c^* \sqrt{n}$ , three*  
778 *structural properties suppress  $\|\eta_{l,c}\|_2$  well below 2. (i) Mean removal is a rank-one correction whose*  
779 *effect on the  $n$ -dimensional unit direction is of order  $|\mu|/\sigma$ , which is empirically small ( $< 0.05$ ) at all*  
780 *cascade levels for well-trained CNN encoders processing natural images. (ii) Variance normalisation*  
781 *(dividing by  $\ell_2$  norm of a zero-mean vector) is direction-preserving. (iii) Affine rescaling within*  
782 *intermediate LACs is a per-channel scalar operation, preserving the zero-mean direction. We*  
783 *empirically observe  $\|R_l\|_\infty / \|\hat{X}_l\|_\infty < 0.03$  across all layers and test images.*

## 784 B.6 Corollary B.6: Equivalence Error Bound

785 **Corollary 2** (B.6: Equivalence Error Bound). *Under the same notation and conditions as Theorem 3,*  
 786 *identifying  $\text{FV}_{l,c}(X)$  with the realised normalised direction  $P_{\perp 1} f_{l,c} / \|P_{\perp 1} f_{l,c}\|_2$  (absorbing the*  
 787 *cascade direction deviation  $\eta_{l,c}$  into the definition of the feature visualisation itself), the reconstruction*  
 788 *error satisfies*

$$\left\| \hat{X}_l - \sum_{c \in \mathcal{C}^+} \tilde{E}_{l,c} \text{FV}_{l,c}(X) \right\|_{\infty} \leq \sum_{c=1}^{C_l} E_{l,c} |\beta_c^*|. \quad (65)$$

789 *In particular, as  $\beta^* \rightarrow 0$  under training convergence (Lemma 1), the bound vanishes and the*  
 790 *feature-visualisation expansion becomes an exact algebraic identity.*

791 *Proof.* Starting from (55), when  $\text{FV}_{l,c}(X)$  is identified with  $P_{\perp 1} f_{l,c} / \|P_{\perp 1} f_{l,c}\|_2$ , the cascade  
 792 direction residual  $\eta_{l,c}$  vanishes by definition. Substituting into the channel superposition:

$$\hat{X}_l = \sum_{c \in \mathcal{C}^+} \tilde{E}_{l,c} \text{FV}_{l,c}(X) + \left( \sum_{c=1}^{C_l} E_{l,c} \beta_c^* \right) \mathbf{1}. \quad (66)$$

793 The residual is a spatially uniform constant whose  $\ell_{\infty}$  norm equals the absolute value of its scalar  
 794 coefficient:

$$\|R_l\|_{\infty} = \left| \sum_{c=1}^{C_l} E_{l,c} \beta_c^* \right| \leq \sum_{c=1}^{C_l} E_{l,c} |\beta_c^*|. \quad (67)$$

795 By Lemma 1(i), this bound vanishes at convergence.  $\square$

796 **Remark 6** (Relationship to Theorem 3). *Theorem 3 defines  $\text{FV}_{l,c}(X)$  as the ideal single-step*  
 797 *gradient direction (without mean-leakage contamination), yielding two residual terms. Corollary 2*  
 798 *defines  $\text{FV}_{l,c}(X)$  as the realised cascade direction, collapsing the decomposition to a single scalar*  
 799 *bias residual. The corollary is therefore tighter as a bound on reconstruction error, while the theorem*  
 800 *provides a more informative decomposition of where the approximation error originates.*

## 801 B.7 Proof of Corollary 1 (Zero Structural Hallucination)

802 We prove Corollary 1 from the main text:  $\text{supp}(\nabla_{i,j} \hat{X}_l) \subseteq \overline{\bigcup_{c \in \mathcal{C}^+} \text{EF}_{l,c}(X)}$ .

803 **Notation.** Throughout this proof,  $\nabla_{i,j}$  denotes the discrete spatial gradient operator: for  $v \in \mathbb{R}^{H_0 \times W_0}$   
 804 at an interior pixel  $(i, j)$ ,

$$(\nabla_{i,j} v)_{(i,j)} := (v_{i+1,j} - v_{i,j}, v_{i,j+1} - v_{i,j}) \in \mathbb{R}^2. \quad (68)$$

805 The spatial gradient support is  $\text{supp}(\nabla_{i,j} v) := \{(i, j) : (\nabla_{i,j} v)_{(i,j)} \neq \mathbf{0}\}$ . For any  $S \subseteq$   
 806  $\{1, \dots, H_0\} \times \{1, \dots, W_0\}$ , the spatial topological closure  $\bar{S}$  used in the main text is the one-pixel  
 807  $(\ell_1)$  dilation

$$\bar{S} \equiv S^{\oplus 1} := \{(i, j) : \exists (i', j') \in S \text{ with } |i - i'| + |j - j'| \leq 1\}. \quad (69)$$

808 Whenever  $\text{supp}(v) \subseteq S$ , one has  $\text{supp}(\nabla_{i,j} v) \subseteq \bar{S}$  by direct inspection of (68).

### 809 Nested Effective Field Condition.

810 **Definition 1** (Nested Effective Field Condition). *The encoder satisfies the nested effective field*  
 811 *condition for layer  $l$  if, for every intermediate level  $k \in \{0, \dots, l-1\}$  and channel  $c' \in \{1, \dots, C_k\}$ ,*

$$\text{EF}_{k,c'}(X) \subseteq \bigcup_{c \in \mathcal{C}^+} \text{EF}_{l,c}(X). \quad (70)$$

812 *Informally: every input pixel that causally influences any active position at any intermediate encoder*  
 813 *level also influences at least one active channel at the target layer  $l$ .*

814 This condition is generically satisfied in fully channel-connected convolutional encoders applied to  
 815 natural images, where active positions at any sufficiently deep level cover the entire input grid.

816 **Cascade Spatial Gradient Containment Lemma.**

817 **Lemma 2** (Cascade Spatial Gradient Containment). *If the encoder satisfies Definition 1 for layer  $l$ ,*  
 818 *then the pre-GroupNorm cascade signal  $f_{l,c}$  defined in (52) satisfies*

$$\text{supp}(f_{l,c}) \subseteq \bigcup_{c' \in \mathcal{C}^+} \text{EF}_{l,c'}(X), \quad (71)$$

819 *and consequently*

$$\text{supp}(\nabla_{i,j} f_{l,c}) \subseteq \left( \bigcup_{c' \in \mathcal{C}^+} \text{EF}_{l,c'}(X) \right)^{\oplus 1}. \quad (72)$$

820 *Proof.* We prove (71); (72) then follows from the dilation property of discrete gradients. Write  
 821  $\mathcal{E}_l := \bigcup_{c' \in \mathcal{C}^+} \text{EF}_{l,c'}(X)$  and proceed by backward induction.

822 *Base case (no intermediate LAC).* Without intermediate LACs, the cascade reduces to  $J_{\text{stem}}^T \circ J_0^T \circ \dots \circ$   
 823  $J_l^T(\text{seed}_{l,c}) = \nabla_X \Phi_{l,c}(X)$  by Proposition 1, so by Theorem 1,  $\text{supp}(\nabla_X \Phi_{l,c}) \subseteq \text{EF}_{l,c}(X) \subseteq \mathcal{E}_l$ .

824 *Inductive step.* At intermediate level  $k$  ( $0 \leq k \leq l-1$ ), the backward signal  $u_k \in \mathbb{R}^{C_k \times H_k \times W_k}$   
 825 enters  $\Psi_{\text{LAC}_k}$ . By Proposition 2,

$$[\Psi_{\text{LAC}_k}(u_k)]_{c'} = a_{k,c'} u_{k,c'} + b_{k,c'} \mathbf{1}, \quad (73)$$

826 with  $a_{k,c'} = \gamma_{c'}^{(k)} / \sigma_{k,c'} > 0$  and  $b_{k,c'} = \beta_{c'}^{(k)} - \gamma_{c'}^{(k)} \mu_{k,c'} / \sigma_{k,c'}$  spatially constant scalars. The signal  
 827 entering the next Jacobian transpose is

$$J_k^T [\Psi_{\text{LAC}_k}(u_k)] = \underbrace{J_k^T(\text{diag}(a_k) u_k)}_{\text{scaled main}} + \underbrace{J_k^T(b_k \otimes \mathbf{1})}_{\text{mean leakage}}. \quad (74)$$

828 *Main signal.*  $\text{diag}(a_k) u_k$  has the same value support as  $u_k$  (positive per-channel rescaling), so  
 829  $J_k^T(\text{diag}(a_k) u_k)$  has value support within the projection of  $\mathcal{E}_l$  to level  $k-1$  by the inductive  
 830 hypothesis.

831 *Mean leakage.*  $J_k^T(b_k \otimes \mathbf{1})(c'', i, j) = \sum_{c'} b_{k,c'} \sum_{r,s} \partial h_{k,c'}(r, s) / \partial h_{k-1,c''}(i, j)$ . By the same  
 832 ReLU gating argument as in Theorem 1, this is nonzero only at positions  $(c'', i, j)$  in the receptive  
 833 field of an active position at level  $k$ . Propagating to input resolution and applying Definition 1:  
 834  $\bigcup_{c'=1}^{C_k} \text{EF}_{k,c'}(X) \subseteq \mathcal{E}_l$ .

835 *Combining.* The value support of  $f_{l,c}$  at input resolution is contained in  $\mathcal{E}_l$ , establishing (71).  $\square$

836 **Proof of Corollary 1.**

837 *Proof. Step 1 (Spatial invariance of the affine bias).* For every channel  $c$  and bias  $\beta_c^* \in \mathbb{R}$ ,  
 838  $\beta_c^* \mathbf{1} \in \mathbb{R}^{H_0 \times W_0}$  is spatially constant, so  $(\nabla_{i,j}(\beta_c^* \mathbf{1}))_{(i,j)} = \mathbf{0}$  for every pixel. This is a pointwise  
 839 algebraic identity, independent of the magnitude of  $\beta_c^*$ .

840 **Step 2 (Mean-centring and scalar division preserve spatial gradient support).** Since  
 841  $[P_{\perp \mathbf{1}} v]_{(i,j)} = v_{(i,j)} - \mu_v$  subtracts a spatial scalar,

$$(\nabla_{i,j}(P_{\perp \mathbf{1}} v))_{(i,j)} = (\nabla_{i,j} v)_{(i,j)} \quad (75)$$

842 pointwise, hence  $\text{supp}(\nabla_{i,j}(P_{\perp \mathbf{1}} v)) = \text{supp}(\nabla_{i,j} v)$ . An analogous identity holds for division by  
 843  $\sigma > 0$ :  $\text{supp}(\nabla_{i,j}(v/\sigma)) = \text{supp}(\nabla_{i,j} v)$ .

844 **Step 3 (Spatial gradient support of  $\hat{X}_l$ ).** Write  $\mathcal{E}_l := \bigcup_{c \in \mathcal{C}^+} \text{EF}_{l,c}(X)$ . By the exact output  
 845 formula (55),

$$\tilde{V}_{l,c} = \gamma_c^* \sqrt{n} d_{l,c}^{\text{cas}} + \beta_c^* \mathbf{1}, \quad d_{l,c}^{\text{cas}} := P_{\perp \mathbf{1}} f_{l,c} / \|P_{\perp \mathbf{1}} f_{l,c}\|_2. \quad (76)$$

846 (3a) Per-channel spatial gradient. Applying  $\nabla_{i,j}$  and using Step 1 (bias drops out) and the positive  
 847 scalar  $\gamma_c^* \sqrt{n} > 0$ :

$$\nabla_{i,j} \tilde{V}_{l,c} = \gamma_c^* \sqrt{n} \nabla_{i,j} d_{l,c}^{\text{cas}}. \quad (77)$$

848 By Step 2 applied to the projection and the scalar division,

$$\text{supp}(\nabla_{i,j} d_{l,c}^{\text{cas}}) = \text{supp}(\nabla_{i,j} f_{l,c}). \quad (78)$$

849 By Lemma 2,  $\text{supp}(\nabla_{i,j} f_{l,c}) \subseteq \mathcal{E}_l^{\oplus 1}$ , so  $\text{supp}(\nabla_{i,j} \tilde{V}_{l,c}) \subseteq \mathcal{E}_l^{\oplus 1}$ .

850 (3b) Channel superposition. By linearity,

$$\nabla_{i,j} \hat{X}_l = \sum_{c=1}^{C_l} E_{l,c} \nabla_{i,j} \tilde{V}_{l,c} = \sum_{c \in \mathcal{C}^+} \tilde{E}_{l,c} \nabla_{i,j} d_{l,c}^{\text{cas}}. \quad (79)$$

851 The support of a finite sum is contained in the union of the individual supports, so

$$\text{supp}(\nabla_{i,j} \hat{X}_l) \subseteq \bigcup_{c \in \mathcal{C}^+} \text{supp}(\nabla_{i,j} d_{l,c}^{\text{cas}}) \subseteq \mathcal{E}_l^{\oplus 1}. \quad (80)$$

852 **Step 4 (Non-emptiness of  $\mathcal{C}^+$ ).** The simplex normalisation  $\sum_c E_{l,c} = 1$  with  $E_{l,c} \geq 0$  forces at  
853 least one  $E_{l,c_0} \geq 1/C_l > 0$ , so  $c_0 \in \mathcal{C}^+ \neq \emptyset$  and  $\mathcal{E}_l^{\oplus 1}$  is non-empty.  $\square$

854 **Remark 7** (Exactness, independent of the residual bound). *Theorem 3 bounds  $\|R_l\|_\infty$  by combining*  
855  *$|\beta_c^*|$  and  $\epsilon_{l,c}^{\text{DC}}$ . The present corollary does not invoke this bound. Instead, it works with the exact*  
856 *per-channel output (76) and exploits the fact that both error sources are structurally incapable of*  
857 *introducing spatial gradients outside  $\mathcal{E}_l^{\oplus 1}$ :  $R_l^\beta$  is spatially constant (zero spatial gradient), and*  
858  *$R_l^{\text{DC}} = \sum_c \tilde{E}_{l,c} \eta_{l,c}$  has both  $d_{l,c}^{\text{cas}}$  and  $\text{FV}_{l,c}(X)$  supported in  $\mathcal{E}_l$ . The spatial gradient support*  
859 *containment thus holds exactly, independent of whether  $\|R_l\|_\infty$  is small.*

860 **Remark 8** (Comparison with GAN and diffusion decoders). *Standard generative decoders (trans-*  
861 *posed convolutions in GANs, iterative denoising in diffusion models) introduce learnable spatial*  
862 *kernels in the decoder pathway whose support is not structurally constrained by the encoder's recep-*  
863 *tive field. Such decoders can produce spatial gradients at arbitrary pixel locations, including regions*  
864 *outside every  $\text{EF}_{l,c}(X)$ . The containment guarantee of Corollary 1 is therefore a distinguishing*  
865 *structural property of the LAC decoder architecture: it holds by algebraic construction, not by*  
866 *empirical regularisation.*

## 867 B.8 Proposition B.8: Path Invariance of Shared LAC Parameters

868 **Setup.** Consider a shared adjoint bridge in which the LAC module  $\Psi_{\text{LAC}_b}$  at level  $b$  processes  
869 gradient signals from every source stage  $l \geq b$ . Write  $n_b := H_b W_b$ . The per-channel signal arriving  
870 from stage  $l$  is  $v_{b,c}^{(l)} \in \mathbb{R}^{H_b \times W_b}$ , assumed not spatially degenerate ( $v_{b,c}^{(l)} \notin \text{span}(\mathbf{1})$ ). Per-channel  
871 mean and standard deviation:

$$\mu_{b,c}^{(l)} = \frac{1}{n_b} \sum_{r,s} v_{b,c}^{(l)}(r,s), \quad \sigma_{b,c}^{(l)} = \sqrt{\frac{1}{n_b} \sum_{r,s} (v_{b,c}^{(l)}(r,s) - \mu_{b,c}^{(l)})^2} > 0. \quad (81)$$

872 Standardised signal and full LAC output:

$$\hat{v}_{b,c}^{(l)}(r,s) = \frac{v_{b,c}^{(l)}(r,s) - \mu_{b,c}^{(l)}}{\sigma_{b,c}^{(l)}}, \quad (82)$$

873

$$o_{b,c}^{(l)}(r,s) = \gamma_c^{(b)} \hat{v}_{b,c}^{(l)}(r,s) + \beta_c^{(b)}, \quad \theta_b := \{\gamma_c^{(b)}, \beta_c^{(b)}\}_{c=1}^{C_b}. \quad (83)$$

874 **Proposition 3** (B.8: Path Invariance of Shared LAC Parameters). *Under the shared adjoint bridge*  
875 *design with per-channel GroupNorm (`num_groups = num_channels`), the following three structural*  
876 *properties hold for all source stages  $l \geq b$ :*

877 (a) **Exact Moment Invariance.**  $\hat{v}_{b,c}^{(l)}$  has zero spatial mean and unit spatial second moment, as  
878 exact algebraic identities independent of  $l$  and  $X$ .

879 (b) **Scalar Insensitivity.** The scalar parameters  $(\gamma_c^{(b)}, \beta_c^{(b)})$  are structurally incapable of differ-  
880 entially responding to spatially localised discrepancies between standardised inputs from  
881 distinct source stages.

882 (c) **Gradient Path Locality.** The gradient path length for  $\theta_b$  is exactly  $(b + 1)$  Jacobian-LAC  
 883 pairs, independent of the source stage  $l$ .

884 *Proof. Part (a): Exact Moment Invariance.* Direct algebraic verification.

885 *Zero spatial mean.*

$$\begin{aligned} \frac{1}{n_b} \sum_{r,s} \hat{v}_{b,c}^{(l)}(r, s) &= \frac{1}{\sigma_{b,c}^{(l)}} \cdot \frac{1}{n_b} \sum_{r,s} [v_{b,c}^{(l)}(r, s) - \mu_{b,c}^{(l)}] \\ &= \frac{1}{\sigma_{b,c}^{(l)}} [\mu_{b,c}^{(l)} - \mu_{b,c}^{(l)}] = 0. \end{aligned} \quad (84)$$

886 *Unit spatial second moment.*

$$\frac{1}{n_b} \sum_{r,s} [\hat{v}_{b,c}^{(l)}(r, s)]^2 = \frac{[\sigma_{b,c}^{(l)}]^2}{[\sigma_{b,c}^{(l)}]^2} = 1. \quad (85)$$

887 Both (84)–(85) are algebraic tautologies, holding pointwise for every  $X$ , every channel, every source  
 888 stage, with no dependence on  $\theta_b$ . Geometrically,  $\hat{v}_{b,c}^{(l)}$  lies on the intersection of the zero-mean  
 889 hyperplane and the sphere of radius  $\sqrt{n_b}$ , an  $(n_b - 2)$ -dimensional manifold identical across source  
 890 stages.

891 **Part (b): Scalar Insensitivity.**

892 *Step 1 (Isotropic action).* The map  $T_{\gamma_c}(u) = \gamma_c u$  has all  $n_b$  singular values equal to  $|\gamma_c|$ ; it is an  
 893 isotropic scaling. For two source stages  $A, B$ ,

$$o_{b,c}^{(A)} - o_{b,c}^{(B)} = \gamma_c^{(b)} [\hat{v}_{b,c}^{(A)} - \hat{v}_{b,c}^{(B)}], \quad (86)$$

894 a uniform rescaling of the input discrepancy. The spatial structure of the discrepancy is preserved  
 895 exactly: not reshaped, not selectively amplified, not suppressed at any location.

896 *Step 2 (Dimensional bottleneck).* Contrast with a spatially varying modulation  $T_w(u) = w \odot u$   
 897 ( $w \in \mathbb{R}^{n_b}$ ) which has  $n_b$  independent degrees of freedom. The scalar  $T_{\gamma_c}$  has exactly one parameter,  
 898 collapsing the full spatial correction space to a one-dimensional subspace. The ratio  $1/(n_b - 1)$   
 899 vanishes as resolution grows, formalising the parametric bottleneck.

900 *Step 3 (Scalar aggregate optimality).* Under the aggregate loss  $\mathcal{L} = \sum_{l \geq b} \mathcal{L}_l$ , the optimal  $\gamma_c^*$  satisfies  
 901 the scalar stationary condition

$$\frac{\partial \mathcal{L}}{\partial \gamma_c^{(b)}} = \sum_{l \geq b} \langle g_{l,c}, \hat{v}_{b,c}^{(l)} \rangle_{n_b} = 0, \quad (87)$$

902 where  $g_{l,c} := \partial \mathcal{L}_l / \partial o_{b,c}^{(l)}$ . This is a single scalar equation in a single scalar unknown; the spatial  
 903 variation of both  $g_{l,c}$  and  $\hat{v}_{b,c}^{(l)}$  is integrated out. By Part (a), all  $\hat{v}_{b,c}^{(l)}$  have the same  $\ell_2$  norm and zero  
 904 mean, so the inner products differ across paths only through residual directional differences which  
 905 the scalar  $\gamma_c$  has no capacity to exploit individually. An identical argument applies to  $\beta_c^{(b)}$ :

$$\frac{\partial \mathcal{L}}{\partial \beta_c^{(b)}} = \sum_{l \geq b} \sum_{r,s} g_{l,c}(r, s) = 0, \quad (88)$$

906 involving only the spatial sum of the downstream gradient, independent of the directional content of  
 907  $\hat{v}_{b,c}^{(l)}$ .

908 **Part (c): Gradient Path Locality.**

909 *Step 1 (Computational graph decomposition).* The forward computation connecting  $\theta_b$  to  $\mathcal{L}_l$   
 910 decomposes into: (i) upstream chain  $\mathcal{U}_{l \rightarrow b}$ , mapping the activation at the source stage to the arriving  
 911 signal  $v_{b,c}^{(l)}$  (depends on  $l$ , no parameter from  $\theta_b$ ); (ii) local affine  $\mathcal{A}_b(\theta_b)$ , parameter-free GroupNorm  
 912 followed by affine  $\theta_b$ ; (iii) downstream chain  $\mathcal{D}_{b \rightarrow 0}$ , identical for all  $l \geq b$ .

913 *Step 2 (Parameter independence of standardised input).*  $\hat{v}_{b,c}^{(l)}$  is computed from  $v_{b,c}^{(l)}$  by parameter-free  
 914 operations. Since  $v_{b,c}^{(l)}$  is produced entirely by the upstream chain (independent of  $\theta_b$ ),

$$\frac{\partial \hat{v}_{b,c}^{(l)}(r, s)}{\partial \theta_b} = 0 \quad \forall (r, s), l \geq b. \quad (89)$$

915  $\hat{v}_{b,c}^{(l)}$  is a leaf node with respect to  $\theta_b$ .

916 *Step 3 (Gradient formulae).* By the chain rule,

$$\frac{\partial \mathcal{L}_l}{\partial \gamma_c^{(b)}} = \langle g_{l,c}, \hat{v}_{b,c}^{(l)} \rangle_{n_b}, \quad (90)$$

$$\frac{\partial \mathcal{L}_l}{\partial \beta_c^{(b)}} = \langle g_{l,c}, \mathbf{1} \rangle_{n_b}. \quad (91)$$

917 The only  $l$ -dependent quantities are  $\hat{v}_{b,c}^{(l)}$  (multiplicative coefficient, not a node through which gradient  
 918 propagates) and  $g_{l,c}$  (entering via the downstream chain).

919 *Step 4 (Path length).*  $g_{l,c}$  is computed by backpropagating through the downstream chain  $\mathcal{D}_{b \rightarrow 0}$ ,  
 920 which contains exactly the LAC and Jacobian operations at levels  $b, b-1, \dots, 0$ , plus the stem.  
 921 This is  $(b+1)$  Jacobian-LAC pairs, determined solely by the architectural depth below level  $b$  and  
 922 independent of  $l$ .  $\square$

923 **Remark 9** (Numerical stability parameter  $\varepsilon$ ). *Standard implementations replace the denominator*  
 924 *in (82) with  $\sqrt{[\sigma_{b,c}^{(l)}]^2 + \varepsilon}$  for small  $\varepsilon > 0$ . The zero-mean identity (84) remains exact. The unit-*  
 925 *second-moment identity acquires a multiplicative correction  $1 - \varepsilon/([\sigma_{b,c}^{(l)}]^2 + \varepsilon) = O(\varepsilon)$  that depends*  
 926 *only on the channel variance, not on the source stage index per se. The path invariance argument is*  
 927 *unaffected.*

928 **Remark 10** (Necessity of `num_groups=num_channels`). *The exact per-channel moment elimination*  
 929 *in Part (a) depends critically on the GroupNorm group structure. Replacing per-channel GroupNorm*  
 930 *with LayerNorm (`num_groups = 1`) makes the spatial mean and variance computed jointly across*  
 931 *all channels; the per-channel moment profiles become path-dependent, destroying the invariance.*  
 932 *BatchNorm, computing statistics across the batch dimension, introduces stochastic path dependence*  
 933 *and at inference uses fixed running statistics, eliminating the exact moment cancellation entirely. The*  
 934 *`num_groups = num_channels` configuration is therefore a structural necessity for path invariance.*

935 **Remark 11** (Contrast with spatially varying modulation). *A spatially varying modulation  $T_w(u) =$*   
 936  *$w \odot u$  with  $w \in \mathbb{R}^{n_b}$  defines a diagonal linear map with  $n_b$  independent parameters and can*  
 937 *selectively amplify the path discrepancy at chosen spatial locations, breaking path invariance. The*  
 938 *scalar affine transform replaces  $w$  by  $\gamma_c \mathbf{1}$ , collapsing  $n_b$  degrees of freedom to one, eliminating*  
 939 *spatial selectivity. This dimensional collapse is the precise mechanism by which the LAC architecture*  
 940 *enforces path invariance through a parametric bottleneck, complementing the moment equalisation*  
 941 *of Part (a).*

## 942 C Additional Experimental Details

### 943 C.1 Reconstruction Fidelity: Qualitative Analysis and Baseline Implementation

944 This appendix expands Section 3.1 of the main text. Figure 1 shows representative outputs from all  
 945 five methods. The qualitative pattern across hundreds of test images is consistent with the quantitative  
 946 scores in Table 1. Vanilla Gradient produces unintelligible high-frequency noise, the direct visual  
 947 signature of the dual pathologies of Section 2.1: zero-insertion spikes have not been corrected, and  
 948 the resulting nail-bed pattern dominates the pixel-space output. Mahendran & Vedaldi recover global  
 949 silhouettes but suffer from severe color distortion and locally hallucinated textures, the typical failure  
 950 mode of  $\text{TV}/\ell_2$  priors that smooth structural discontinuities while introducing prior-induced patterns  
 951 absent from the encoder’s response. Robust Inversion exhibits similar issues, with the additional  
 952 artifact that the adversarially robust encoder itself shifts the response distribution away from typical  
 953 pretrained features. UpConvNet eliminates high-frequency artifacts but at the cost of severe over-  
 954 smoothing of all fine structures, including discriminative edges that the encoder demonstrably uses

(as evidenced by its insertion AUC superiority on the same backbone). Our method alone preserves both global semantic layout and sharp local spatial details, naturalistic colors, and high-frequency textures, consistent with the zero-hallucination guarantee of Corollary 1.

**Baseline implementations.** All baselines are evaluated using their official public implementations on the same frozen encoders. For Mahendran & Vedaldi we follow the natural-image prior formulation of [18], which combines a reconstruction objective with total-variation and  $\ell_2$  regularization terms; we adopt the default regularization weights and optimization schedule reported in the original paper. For Robust Inversion we use the publicly released adversarially-robust encoder weights from [6]. For UpConvNet we train a separate feedforward decoder per backbone-dataset pair following the protocol of [4], training each decoder until validation reconstruction loss plateaus. Our method uses the single LAC cascade trained per backbone (Section 2.4) and applied to all five datasets without retraining. We emphasize that, in contrast to the regularized objectives of all three optimization-based and feedforward baselines, our method’s training loss consists solely of the  $L_1$  pixel reconstruction term (9) with no spatial smoothing or natural-image prior, which is precisely the property that enables the zero-hallucination guarantee of Corollary 1.

## C.2 Foreground Energy Paradox: Setup and Numerical Detail

This appendix expands Section 3.2.1. We evaluate on two fine-grained benchmarks providing pixel-accurate foreground masks: **CUB-200-2011** [27] (200 bird species, 11,788 images) and **Oxford-IIIT Pet** [23] (37 cat/dog breeds, 7,349 images). We test four frozen ImageNet-pretrained encoders: **ResNet-18**, **ResNet-50** [10], **DenseNet-121** [12], and **ConvNeXt-Base** [17]. For each encoder we freeze all convolutional parameters and train a single linear probe  $w^{(c)}$  on the target dataset using standard supervised classification protocol, so that the classifier directions under study are exactly those used for classification. The class index  $c$  in the inversion is taken to be the classifier’s top-1 predicted label. For each test image we compute the three reconstructions  $\hat{X}^{(c)}$ ,  $\hat{X}_+^{(c)}$ ,  $\hat{X}_-^{(c)}$ , evaluate FG on each, and average across the full test split.

**Metric details.** The foreground energy proportion

$$\text{FG}(\hat{X}) = \frac{\sum_{(i,j) \in M_{\text{fg}}} \sum_{k=1}^3 \hat{X}_k(i,j)^2}{\sum_{(i,j)} \sum_{k=1}^3 \hat{X}_k(i,j)^2} \quad (92)$$

sums over the three RGB channels in both numerator and denominator, with no mean-centering applied. The metric thus operates on the raw reconstruction as produced by the framework. A value substantially above the area ratio  $|M_{\text{fg}}|/(H_0 W_0)$  indicates that the reconstruction concentrates its energetic mass on the object beyond what would be expected from uniform energy distribution. On both datasets the foreground area ratio is approximately 0.40–0.55, so values of  $\text{FG} \in [0.55, 0.70]$  for our Full reconstructions correspond to genuine foreground concentration.

## C.3 Visual Falsification of SFH: Detailed Discussion

This appendix expands Section ?? . The Spatial Funnel Hypothesis, defined in Section 1, predicts two structural consequences that our framework allows us to test directly at the pixel level.

**Detailed falsification of P1 (channel sparsification).** Figure 3(a) visualizes  $\tilde{V}_{L-1,i}(X)$  for channels drawn from the high-, mid-, and low-energy groups of the forward simplex measure  $E_{L-1,c}$ . The outcome is uniform across every image and every energy rank we examined: every single channel recovers a visually complete, globally-supported rendering of the input scene, preserving foreground object, contextual surround, and overall spatial layout simultaneously. There is no channel that specializes in a localized fragment of the image; there is no channel that renders only the object while suppressing the background; there is no observable qualitative divide between the high-energy and low-energy groups.

What varies across channels is a global semantic “lens” rather than a spatial support. Different channels emphasize different texture frequencies, chromatic biases, and edge polarities, but all of them render the entire scene. This is the visual fingerprint of a strong superposition code: each channel is a holistic basis vector in pixel space, not a spatially selective filter. The pattern is categorically incompatible with P1 and is consistent across all four backbones we tested.

1003 **Detailed falsification of P2 (sign-hemisphere specialization).** Figure 3(b) juxtaposes, for each  
 1004 input image, the stage-wise reconstructions  $\hat{X}_l$  ( $l \in \{0, 1, 2, 3\}$ ), the class-directional reconstruction  
 1005  $\hat{X}^{(c)}$ , and its two sign-hemisphere components  $\hat{X}_+^{(c)}$  and  $\hat{X}_-^{(c)}$ . Two observations emerge.

1006 First, the stage-wise column reveals that background pixels are never spatially discarded: even the  
 1007 deepest stage  $\hat{X}_3$  retains the full spatial extent of the scene, with every texture, edge, and contextual  
 1008 element still present (albeit reshaped through progressively more abstract semantic lenses). SFH  
 1009 predicts the reconstruction should progressively concentrate on the object as  $l$  increases; instead, what  
 1010 shifts across stages is the style and frequency content of the full-scene rendering, not its spatial support.  
 1011 A typical  $\hat{X}_0$  exhibits photographic-quality reconstruction with rich high-frequency detail;  $\hat{X}_3$  exhibits  
 1012 a more abstracted, lower-frequency rendering with subtle attenuation of non-discriminative regions,  
 1013 but the global scene layout is unmistakably preserved.

1014 Second, and decisively, the positive-hemisphere image  $\hat{X}_+^{(c)}$  and the negative-hemisphere image  $\hat{X}_-^{(c)}$   
 1015 are visually nearly indistinguishable. Both contain the foreground object in the same location, at  
 1016 the same scale, with the same pose; both contain the background in the same layout, with the same  
 1017 textures and the same contextual clutter. Under side-by-side inspection, it is in practice impossible  
 1018 to tell which image corresponds to which hemisphere. Yet the algebraic sum  $\hat{X}_+^{(c)} + \hat{X}_-^{(c)} = \hat{X}^{(c)}$   
 1019 exhibits a qualitatively different spatial distribution, with the background collapsing and a foreground-  
 1020 concentrated silhouette emerging, consistent with the foreground-energy ratios of Section 3.2.1.  
 1021 This pattern is the precise visual counterpart of the numerical signature  $\text{FG}(\hat{X}_+^{(c)}) \approx \text{FG}(\hat{X}_-^{(c)}) \ll$   
 1022  $\text{FG}(\hat{X}^{(c)})$ .

1023 **Double falsification and its implication.** Both P1 and P2 are falsified along their structural  
 1024 predictions. Channels do not sparsify; sign-hemispheres do not specialize. The background of the  
 1025 input is not removed by the encoder, not removed by the positive classifier weights, not removed by  
 1026 the negative classifier weights, and yet vanishes in the algebraic sum of the two. This is the empirical  
 1027 signature not of filtering but of cancellation, and it demands a mechanistic account in which the  
 1028 classifier’s weights serve a fundamentally different role than selection. Section 3.2.2 provides such  
 1029 an account.

#### 1030 C.4 Hypotheses H1 and H2: Empirical Verification and Resolution

1031 This appendix expands Section 3.2.2. We provide the full theoretical motivation for H1, the detailed  
 1032 empirical verification, the H2 derivation, and the complete resolution of the paradox.

1033 **Why a classification-trained encoder retains the background.** Hypothesis H1 posits that every  
 1034 deep channel inversion contains a shared low-rank background component  $C_i(X) \cdot B(X)$  that is  
 1035 common across all channels. We give an information-theoretic justification. The encoder is optimized  
 1036 under classification-only pressure, which depends only on the relations among channels rather than  
 1037 on absolute magnitudes. Concretely, the cross-entropy gradient with respect to  $h_{l,c}(X)$  depends on  
 1038  $\text{logit}_c - \text{logit}_{c'}$  for various  $c, c'$ , all of which are linear in the channel features. Any common-mode  
 1039 component  $B(X)$  that is added with the same scalar coefficient  $C_i \equiv C$  to every channel cancels  
 1040 exactly from each pairwise logit difference, producing zero gradient pressure. The encoder therefore  
 1041 has no incentive to erase content that is common-mode across channels: such content costs nothing  
 1042 under the training objective while preserving representational generality for unseen downstream tasks.  
 1043 This principle is consistent with the strong superposition hypothesis of [5], originally formulated  
 1044 for language models. To our knowledge, Eq. (13) constitutes the first direct pixel-level confirmation  
 1045 of this hypothesis in the vision domain, obtained through an inversion operator with an algebraic  
 1046 zero-hallucination guarantee rather than through activation-statistics inference.

1047 **Detailed empirical verification of H1.** H1 asserts two concrete, testable properties of the channel  
 1048 inversion family  $\{\tilde{V}_{L-1,i}(X)\}_{i=1}^{C_{L-1}}$ : the existence of a dominant shared direction  $B(X)$  in the per-  
 1049 image Gram structure, and the emergence of channel-specific diversity after its removal. We verify  
 1050 both directly.

1051 Let  $G(X) \in \mathbb{R}^{C_{L-1} \times C_{L-1}}$  denote the per-image Gram matrix with entries  $G(X)_{ij} =$   
 1052  $\langle \tilde{V}_{L-1,i}(X), \tilde{V}_{L-1,j}(X) \rangle$ . Eigendecomposition of  $G(X)$  averaged over the evaluation set shows

that the leading eigenvector alone captures 20.9% of the total Gram energy, an order-of-magnitude concentration that is incompatible with a generic isotropic channel ensemble and constitutes direct numerical evidence for the shared direction  $B(X)$  posited in Eq. (13).

Furthermore, rank-one subtraction of this dominant component from each channel (i.e., replacing  $\tilde{V}_{L-1,i}$  by  $\delta_i(X) = \tilde{V}_{L-1,i} - C_i(X)B(X)$ , where  $B(X)$  is the leading eigenvector and  $C_i(X) = \langle \tilde{V}_{L-1,i}, B(X) \rangle$ ) expands the inter-channel variance by a factor of approximately  $1.27 \times 10^4$ , from 0.0001% of the average channel energy to 1.27%. Correspondingly, the distribution of pairwise cosine similarities widens from the extremely narrow range  $[0.18, 0.21]$  to the much broader range  $[-0.07, 0.67]$ . The channel family, which appears nearly isotropic before rank-one subtraction, reveals a richly diverse residual geometry after it. Both observations are precisely the signatures predicted by H1 and are consistent across all four architectures evaluated in Section 3.2.1.

**H2 derivation in full.** Recall the exact adjoint-linearity identity from Section 3.2.1,

$$\nabla_X \sum_i w_i^{(c)} h_i(X) \equiv \sum_i w_i^{(c)} \nabla_X h_i(X), \quad (93)$$

which lifts any scalar classifier operation in logit space to an algebraically equivalent geometric operation in pixel space. Substituting H1 into the class-directional reconstruction yields the exact decomposition

$$\hat{X}^{(c)}(X) = \sum_i w_i^{(c)} \tilde{V}_{L-1,i}(X) = B(X) \underbrace{\sum_i w_i^{(c)} C_i(X)}_{\alpha^{(c)}(X)} + \underbrace{\sum_i w_i^{(c)} \delta_i(X)}_{\text{foreground residual}}. \quad (94)$$

The pixel-space image of any class direction decomposes exactly into a scalar multiple of the shared background  $B(X)$  and a weighted sum of the channel-specific residuals. The numerical behavior of a trained classifier is therefore fully characterized by two coefficient regimes:  $\{w_i^{(c)} C_i(X)\}_i$  controls the magnitude of the background contribution, and  $\{w_i^{(c)} \delta_i(X)\}_i$  controls the foreground content.

A well-trained classifier is precisely one whose weights  $w^{(c)}$  simultaneously satisfy two algebraic boundary conditions:

$$(B1) \quad \alpha^{(c)}(X) = \sum_i w_i^{(c)} C_i(X) \approx 0, \quad (B2) \quad \sum_i w_i^{(c)} \delta_i(X) \text{ aligns on the class-}c \text{ object.} \quad (95)$$

Condition (B1) is a destructive-interference constraint: across the sign-mixed ensemble  $\{w_i^{(c)}\}_i$ , the positive and negative contributions to the background coefficient  $\alpha^{(c)}$  must cancel in expectation over natural images, so that the shared scene content  $B(X)$  is annihilated in the pixel-space sum. Condition (B2) is a constructive-interference constraint: the residuals  $\delta_i(X)$  must superpose coherently on the discriminative structure of class  $c$ . The classifier's scalar weight assignment serves as the set of interference coefficients that simultaneously enforce background annihilation and foreground emergence on the reconstructed pixel manifold.

**Resolution of the paradox.** Hypotheses H1 and H2 together predict, exactly and without further assumption, the empirical signature established in Sections 3.2.1–??.

The indistinguishability of  $\hat{X}_+^{(c)}$  and  $\hat{X}_-^{(c)}$  follows immediately from H1: because every channel carries a full copy of  $B(X)$  with coefficient  $C_i(X)$  of comparable magnitude (regardless of the sign of  $w_i^{(c)}$ ), each sign-restricted partial sum

$$\hat{X}_\pm^{(c)} = B(X) \sum_{i: \text{sign}(w_i^{(c)})=\pm} w_i^{(c)} C_i(X) + \sum_{i: \text{sign}(w_i^{(c)})=\pm} w_i^{(c)} \delta_i(X) \quad (96)$$

retains the background component at full amplitude. The foreground-energy ratios of the two hemispheres are therefore dominated by  $B(X)$  and coincide, as observed (O1).

The sharp jump  $\text{FG}(\hat{X}_\pm^{(c)}) \ll \text{FG}(\hat{X}^{(c)})$  upon summation follows immediately from H2: the sum over all channels satisfies the destructive-interference boundary condition (B1), so the background

coefficient  $\alpha^{(c)}(X)$  collapses toward zero and the scene content is algebraically annihilated. The surviving signal is the foreground residual sum of (B2), whose energy is concentrated on the class-specific object by training (O2).

The Spatial Funnel Hypothesis is thus replaced by a different physical picture: background pixels are never spatially discarded; they are algebraically cancelled in the pixel-space superposition that the classifier induces. The encoder is a holographic analyzer; the classifier is a precisely-tuned linear interferometer; classification, at the level of the pixel manifold, is literally an interference experiment whose coefficients have been solved for by gradient descent. This reinterpretation does not contradict the statistical effectiveness of classification or the empirical utility of CAM-style visualizations; it reframes their geometric substrate.

## C.5 Proof of Theorem 2 (Covariance-Volume Duality)

This appendix expands Section 3.2.3 with the full proof, the formal volume definition, and the GAP-space optimization rationale.

**Formal definition of the parallelepiped volume.** The admissible interference subspace  $\mathcal{F}_S(X)$  is a  $(|S| - 1)$ -dimensional linear subspace of  $\mathbb{R}^{3H_0W_0}$ . Its functional capacity is measured by the parallelepiped volume of the image of any orthonormal basis of the constraint hyperplane  $\{w : C_S(X)^\top w = 0\} \subset \mathbb{R}^{|S|}$  under the linear map  $\Delta_S(X) = [\delta_i(X)]_{i \in S}$ :

$$\text{Vol}(\mathcal{F}_S(X)) := \sqrt{\det(U(X)^\top \Delta_S(X)^\top \Delta_S(X) U(X))}, \quad (97)$$

where  $U(X) \in \mathbb{R}^{|S| \times (|S|-1)}$  is any orthonormal basis of  $C_S(X)^\perp$ . By the standard invariance of the Gram determinant under change of orthonormal basis, the right-hand side does not depend on the particular choice of  $U(X)$  and is therefore an intrinsic geometric invariant of the subspace.

**Theorem statement.** Let  $H \in \mathbb{R}^{N \times C_L-1}$  denote the GAP feature matrix of a dataset of  $N$  images, with  $H_{n,i} = h_i(X_n)$ , and let  $\Sigma_S$  denote the centered covariance submatrix indexed by  $S$ . Assume Hypothesis H1 holds and the shared background direction  $B(X)$  is not exactly aligned with any residual subspace  $\text{span}(\{\delta_i(X)\}_{i \in S})$ . Then

$$\arg \max_{|S|=k} \mathbb{E}_X [\log \text{Vol}^2(\mathcal{F}_S(X))] = \arg \max_{|S|=k} \log \det(\Sigma_S). \quad (98)$$

*Proof.* The argument proceeds in four steps.

*Step 1 (Structural expansion of  $\mathcal{F}_S(X)$ ).* Expanding each  $\tilde{V}_{L-1,i}(X)$  via H1,

$$\sum_{i \in S} w_i \tilde{V}_{L-1,i}(X) = \left( \sum_{i \in S} w_i C_i(X) \right) B(X) + \sum_{i \in S} w_i \delta_i(X). \quad (99)$$

The constraint  $C_S(X)^\top w = 0$  annihilates the first term, so  $\mathcal{F}_S(X) = \{\Delta_S(X)w : C_S(X)^\top w = 0\}$ , the image under  $\Delta_S(X)$  of the constraint hyperplane.

*Step 2 (Reduction to a Gram determinant).* By the volume definition (97),

$$\text{Vol}^2(\mathcal{F}_S(X)) = \det(U^\top G_\delta U), \quad G_\delta(X) := \Delta_S(X)^\top \Delta_S(X). \quad (100)$$

Using the standard identity for the determinant of an orthogonal restriction of a positive-definite matrix,

$$\det(U^\top G_\delta U) = \det(G_\delta) \cdot \frac{C_S^\top G_\delta^{-1} C_S}{C_S^\top C_S}, \quad (101)$$

valid whenever  $G_\delta$  is nonsingular. The second factor  $\rho_S(X) := C_S^\top G_\delta^{-1} C_S / \|C_S\|_2^2$  is bounded by  $\lambda_{\max}(G_\delta)^{-1} \leq \rho_S(X) \leq \lambda_{\min}(G_\delta)^{-1}$ . Taking logarithms,

$$\log \text{Vol}^2(\mathcal{F}_S(X)) = \log \det(G_\delta(X)) + \log \rho_S(X). \quad (102)$$

*Step 3 (Expectation over the data distribution).* Taking expectations,

$$\mathbb{E}_X [\log \text{Vol}^2(\mathcal{F}_S(X))] = \mathbb{E}_X [\log \det(G_\delta(X))] + \mathbb{E}_X [\log \rho_S(X)]. \quad (103)$$

1124 By concavity of  $\log \det$  on PSD matrices and Jensen's inequality,  $\mathbb{E}_X[\log \det(G_\delta(X))] \leq$   
 1125  $\log \det(\bar{G}_{\delta,S})$  where  $\bar{G}_{\delta,S} := \mathbb{E}_X[\Delta_S(X)^\top \Delta_S(X)]$ . The second term  $\mathbb{E}_X[\log \rho_S(X)]$  is a logarithm  
 1126 of a bounded positive ratio and contributes a correction subdominant to  $\log \det(\bar{G}_{\delta,S})$  whenever the  
 1127 spectrum of  $G_\delta$  is bounded away from degeneracy, a condition directly verified by the rank-one-  
 1128 subtracted variance expansion (1.27% across channels, with no observed degeneracy) reported in  
 1129 Section 3.2.2. Consequently,

$$\arg \max_{|S|=k} \mathbb{E}_X[\log \text{Vol}^2(\mathcal{F}_S(X))] = \arg \max_{|S|=k} \log \det(\bar{G}_{\delta,S}). \quad (104)$$

1130 *Step 4 (Duality between  $\bar{G}_{\delta,S}$  and  $\Sigma_S$ ).* We show  $\bar{G}_{\delta,S}$  equals  $\Sigma_S$  up to a positive diagonal rescaling  
 1131 independent of  $S$ , plus a rank-one correction that does not alter subset rankings. From the LAC  
 1132 additive synthesis  $X = \sum_i h_i(X) \tilde{V}_{L-1,i}(X)$  (Section 2.4), taking inner products with  $\tilde{V}_{L-1,j}(X)$ ,

$$\langle X, \tilde{V}_{L-1,j}(X) \rangle = \sum_i h_i(X) \langle \tilde{V}_{L-1,i}, \tilde{V}_{L-1,j} \rangle. \quad (105)$$

1133 Under H1,  $\langle \tilde{V}_{L-1,i}, \tilde{V}_{L-1,j} \rangle = C_i C_j \|B\|^2 + \langle \delta_i, \delta_j \rangle$ . Taking the empirical covariance of both sides  
 1134 across the dataset, the rank-one term contributes a single common-mode eigen-direction (captured  
 1135 empirically by the 20.9%-energy leading principal component) while the residual term contributes  
 1136  $\bar{G}_{\delta,S}$ . Using the LAC normalization identity  $\|\tilde{V}_{L-1,i}(X)\|_2 = \gamma_i^* \sqrt{H_0 W_0}$  (Proposition 2),

$$\bar{G}_{\delta,S} = D_S \Sigma_S D_S + \text{rank-1 correction}, \quad (106)$$

1137 with  $D_S = \text{diag}(\gamma_i^* \sqrt{H_0 W_0})_{i \in S}$ . The rank-one correction shifts  $\log \det$  by a subset-dependent  
 1138 additive constant bounded by  $\log(1 + \lambda_{\max}/\lambda_{\min})$ , uniformly controlled under the non-degeneracy  
 1139 assumption. Since  $\det(D_S \Sigma_S D_S) = \det(\Sigma_S) \prod_{i \in S} (\gamma_i^*)^2 H_0 W_0$  and the second factor is a product  
 1140 of channel-wise constants, the maximizers of  $\log \det(\bar{G}_{\delta,S})$  and  $\log \det(\Sigma_S)$  coincide whenever the  
 1141 channel-wise scales  $\gamma_i^*$  are approximately uniform, a condition enforced by the shared LAC training  
 1142 (Proposition 3). This yields the duality (98).  $\square$

1143 **Why optimize in GAP space rather than inversion space.** Although Theorem 2 establishes  
 1144 algebraic equivalence between the two domains, only one is numerically well-conditioned. The LAC  
 1145 enforces, by Proposition 2, an exact per-channel normalization  $\|\tilde{V}_{L-1,i}(X)\|_2 = \gamma_i^* \sqrt{H_0 W_0}$  whose  
 1146 magnitude is input-independent. In our experiments this predicted identity is verified with near-zero  
 1147 deviation: across all channels we measure the inversion norm at  $172.24 \pm 0.00$ , with a resulting  
 1148 inter-channel covariance variation of only 0.0001% of the mean channel energy. Direct determinant  
 1149 maximization on the inversion-space Gram matrix is therefore numerically degenerate: the diagonal  
 1150 is constant to machine precision, and partial Cholesky pivoting cannot meaningfully distinguish  
 1151 channels. The GAP feature space is not subject to this constraint: the scalar features  $h_i(X)$  retain  
 1152 their natural per-channel variance structure because GAP is applied on the forward-path activations,  
 1153 which the LAC does not touch. The duality theorem reconciles geometric interpretability (inversion  
 1154 space) with numerical tractability (GAP space): we interpret in one and optimize in the other.

1155 **Physical interpretation of the greedy rule.** Each iteration of Algorithm 1 admits a transparent  
 1156 interpretation in the language of Section 3.2.2. Step 4 selects the channel with the largest residual  
 1157 diagonal entry of the current Schur complement, equivalently, the channel whose GAP feature carries  
 1158 the most variance not yet explained by previously chosen channels. Step 5 then subtracts the subspace  
 1159 spanned by the newly selected channel from all remaining covariance entries, forcing subsequent  
 1160 iterations to seek channels whose residual information is linearly independent of the already-selected  
 1161 set.

1162 The algorithm balances two competing criteria: each channel's individual informativeness (captured  
 1163 by the diagonal magnitude, analogous to the length of an ellipsoid axis) and the mutual independence  
 1164 of the selected set (captured by the Schur-complement reduction, analogous to axis orthogonality).  
 1165 These are precisely the two geometric conditions identified by H2 as necessary for effective destructive  
 1166 interference: individual variance supplies the amplitude required for background cancellation in (B1),  
 1167 and mutual independence supplies the directional diversity required for foreground reconstruction  
 1168 in (B2). The greedy rule is thus not a generic heuristic, it is the algorithmic realization, under the  
 1169 approximation granted by submodularity, of the interference geometry that classification itself relies  
 1170 upon.

## 1171 C.6 OOD Validation: Numerical Tables and Detailed Analysis

1172 This appendix expands Section 3.2.4 with full numerical results and detailed analysis of the three  
1173 findings.

1174 **Full numerical tables.** Tables 4, 6, 8, and 10 report CIFAR-100 (in-distribution) and CIFAR-100-C  
1175 (corruption-OOD) accuracy across all pruning ratios and selection methods for ConvNeXt-Base,  
1176 ResNet-18, ResNet-50, and DenseNet-121, respectively. Tables 5, 7, 9, and 11 report cross-dataset  
1177 transfer results on Stanford Dogs and Oxford-IIIT Pet. All experiments use the same frozen models.

Table 4: ConvNeXt-Base: In-distribution and corruption-OOD accuracy.

| Prune% | CIFAR-100 (ID) |        |            | CIFAR-100-C (OOD) |        |            |
|--------|----------------|--------|------------|-------------------|--------|------------|
|        | CovVol         | Energy | Random     | CovVol            | Energy | Random     |
| 0      | 88.23          | 88.23  | 88.23±0.00 | 62.77             | 62.77  | 62.77±0.00 |
| 50     | 88.12          | 88.20  | 88.19±0.02 | 62.60             | 62.60  | 62.60±0.09 |
| 80     | 88.01          | 87.91  | 88.02±0.12 | 62.04             | 62.08  | 62.08±0.15 |
| 90     | 87.50          | 87.44  | 87.51±0.12 | 61.23             | 61.18  | 61.18±0.20 |
| 95     | 86.61          | 86.32  | 86.26±0.33 | 59.44             | 59.03  | 59.03±0.37 |
| 98     | <b>82.20</b>   | 80.17  | 80.72±0.46 | <b>51.96</b>      | 50.99  | 50.99±0.63 |

Table 5: ConvNeXt-Base: Cross-dataset transfer on Stanford Dogs and Oxford-IIIT Pet.

| Prune% | Stanford Dogs |        |            | Oxford-IIIT Pet |        |            |
|--------|---------------|--------|------------|-----------------|--------|------------|
|        | CovVol        | Energy | Random     | CovVol          | Energy | Random     |
| 0      | 91.75         | 91.75  | 91.75±0.00 | 94.39           | 94.39  | 94.39±0.00 |
| 50     | 91.99         | 91.92  | 91.72±0.15 | 94.03           | 94.25  | 94.43±0.20 |
| 80     | 91.74         | 91.67  | 91.54±0.13 | 93.79           | 93.89  | 93.81±0.18 |
| 90     | 91.36         | 91.31  | 91.18±0.12 | 93.46           | 93.38  | 93.12±0.12 |
| 95     | 90.63         | 90.49  | 90.00±0.14 | 91.53           | 91.59  | 91.47±0.33 |
| 98     | <b>86.93</b>  | 85.49  | 85.12±0.11 | <b>86.43</b>    | 85.09  | 85.31±0.53 |

Table 6: ResNet-18: In-distribution and corruption-OOD accuracy.

| Prune% | CIFAR-100 (ID) |        |            | CIFAR-100-C (OOD) |        |            |
|--------|----------------|--------|------------|-------------------|--------|------------|
|        | CovVol         | Energy | Random     | CovVol            | Energy | Random     |
| 0      | 78.09          | 78.09  | 78.09±0.00 | 49.40             | 49.40  | 49.40±0.00 |
| 50     | 77.88          | 77.68  | 77.80±0.15 | 49.14             | 49.01  | 49.12±0.04 |
| 80     | 77.20          | 76.69  | 76.98±0.17 | 48.43             | 47.92  | 48.11±0.09 |
| 90     | 75.43          | 75.29  | 75.11±0.19 | 46.71             | 45.92  | 46.10±0.22 |
| 95     | 67.96          | 66.62  | 67.25±0.42 | 38.57             | 37.48  | 38.16±0.27 |
| 98     | <b>50.79</b>   | 49.59  | 49.53±0.51 | <b>26.59</b>      | 25.46  | 25.87±0.31 |

Table 7: ResNet-18: Cross-dataset transfer on Stanford Dogs and Oxford-IIIT Pet.

| Prune% | Stanford Dogs |        |            | Oxford-IIIT Pet |        |            |
|--------|---------------|--------|------------|-----------------|--------|------------|
|        | CovVol        | Energy | Random     | CovVol          | Energy | Random     |
| 0      | 78.62         | 78.62  | 78.62±0.00 | 89.32           | 89.32  | 89.32±0.00 |
| 50     | 77.28         | 76.66  | 76.93±0.29 | 87.16           | 86.94  | 87.64±0.52 |
| 80     | 73.26         | 72.44  | 73.35±0.40 | 83.32           | 83.21  | 84.25±0.31 |
| 90     | 68.08         | 67.55  | 65.30±1.91 | 77.57           | 79.72  | 78.64±1.07 |
| 95     | 52.76         | 47.89  | 42.81±2.69 | 68.30           | 64.90  | 60.41±3.61 |
| 98     | <b>25.99</b>  | 23.82  | 22.49±2.88 | <b>45.54</b>    | 43.50  | 40.39±3.08 |

1178 **Detailed Finding 1: a robustness plateau refutes spatial specialization.** Below approximately  
1179 90% pruning, all three selection methods perform within fractions of a percent of full-model baselines,  
1180 with random selection statistically indistinguishable from the principled selectors. Under SFH, deep

Table 8: ResNet-50: In-distribution and corruption-OOD accuracy.

| Prune% | CIFAR-100 (ID) |        |            | CIFAR-100-C (OOD) |        |            |
|--------|----------------|--------|------------|-------------------|--------|------------|
|        | CovVol         | Energy | Random     | CovVol            | Energy | Random     |
| 0      | 79.23          | 79.23  | 79.23±0.00 | 49.83             | 49.83  | 49.83±0.00 |
| 50     | 79.24          | 79.18  | 79.16±0.05 | 49.80             | 49.74  | 49.80±0.03 |
| 80     | 79.38          | 79.26  | 79.10±0.11 | 49.81             | 49.46  | 49.68±0.07 |
| 90     | 78.83          | 78.60  | 78.61±0.15 | 49.54             | 49.04  | 49.32±0.12 |
| 95     | 77.58          | 77.12  | 77.24±0.15 | 48.19             | 47.28  | 47.51±0.28 |
| 98     | <b>75.55</b>   | 74.22  | 75.14±0.26 | <b>45.43</b>      | 44.16  | 45.04±0.35 |

Table 9: ResNet-50: Cross-dataset transfer on Stanford Dogs and Oxford-IIIT Pet.

| Prune% | Stanford Dogs |        |            | Oxford-IIIT Pet |        |            |
|--------|---------------|--------|------------|-----------------|--------|------------|
|        | CovVol        | Energy | Random     | CovVol          | Energy | Random     |
| 0      | 89.34         | 89.34  | 89.34±0.00 | 90.76           | 90.76  | 90.76±0.00 |
| 50     | 88.64         | 89.04  | 88.59±0.10 | 89.42           | 89.72  | 90.20±0.17 |
| 80     | 87.38         | 87.11  | 86.55±0.28 | 88.39           | 87.98  | 88.15±0.32 |
| 90     | 85.29         | 85.05  | 84.48±0.40 | 85.80           | 85.72  | 85.39±0.56 |
| 95     | 84.03         | 83.20  | 73.84±0.46 | 82.86           | 81.95  | 79.22±1.37 |
| 98     | <b>81.18</b>  | 80.05  | 54.93±3.01 | <b>80.32</b>    | 78.74  | 66.31±3.43 |

Table 10: DenseNet-121: In-distribution and corruption-OOD accuracy.

| Prune% | CIFAR-100 (ID) |        |            | CIFAR-100-C (OOD) |        |            |
|--------|----------------|--------|------------|-------------------|--------|------------|
|        | CovVol         | Energy | Random     | CovVol            | Energy | Random     |
| 0      | 82.60          | 82.60  | 82.60±0.00 | 53.32             | 53.32  | 53.32±0.00 |
| 50     | 82.53          | 82.53  | 82.35±0.13 | 53.41             | 53.41  | 53.05±0.13 |
| 80     | 81.92          | 82.10  | 81.78±0.22 | 52.79             | 52.58  | 51.93±0.15 |
| 90     | 81.30          | 80.90  | 80.02±0.34 | 51.50             | 51.08  | 49.80±0.32 |
| 95     | 78.22          | 77.26  | 74.19±0.98 | 47.50             | 46.44  | 43.49±0.97 |
| 98     | <b>70.00</b>   | 67.19  | 59.59±3.32 | <b>39.92</b>      | 36.74  | 32.36±2.23 |

Table 11: DenseNet-121: Cross-dataset transfer on Stanford Dogs and Oxford-IIIT Pet.

| Prune% | Stanford Dogs |        |            | Oxford-IIIT Pet |        |            |
|--------|---------------|--------|------------|-----------------|--------|------------|
|        | CovVol        | Energy | Random     | CovVol          | Energy | Random     |
| 0      | 78.99         | 78.99  | 78.99±0.00 | 88.58           | 88.58  | 88.58±0.00 |
| 50     | 79.14         | 77.47  | 76.02±0.69 | 88.42           | 88.14  | 87.29±0.52 |
| 80     | 77.18         | 73.60  | 68.00±1.54 | 85.17           | 86.02  | 81.89±1.27 |
| 90     | 71.62         | 67.00  | 57.63±1.25 | 81.79           | 80.98  | 74.68±1.36 |
| 95     | 56.62         | 50.02  | 35.79±1.99 | 71.74           | 68.96  | 56.69±3.05 |
| 98     | <b>36.17</b>  | 31.34  | 21.39±4.36 | <b>57.32</b>    | 50.83  | 41.30±4.94 |

convolutional channels would be specialized to spatially localized, class-discriminative fragments, and random discarding of 90% of them would produce catastrophic accuracy collapse. The observed plateau is the precise signature expected under holographic superposition (H1): because every channel already carries a full copy of the input scene, most channels are informationally redundant, and the classifier can reconstruct the destructive-interference geometry from almost any sufficiently large subset of them. This finding is independent corroboration of H1 distinct from the FG paradox of Section 3.2.1.

**Detailed Finding 2: in the tail, the interference geometry reveals itself.** The three methods separate only at extreme compression ( $\rho \geq 95\%$ ), where the channel budget approaches the dimensionality of the task-discriminative residual subspace. CovVol pulls systematically ahead: at 98% pruning on CIFAR-100 it achieves 82.20% vs. 80.17% (Energy) and 80.72% (Random); on CIFAR-100-C the gap widens to 51.96 vs. 48.98 vs. 50.99. The relative CovVol-vs-Energy gap is uniformly larger on OOD than in-distribution, indicating that covariance-volume selection preserves

underlying interference capacity rather than merely training-distribution accuracy. This finding directly corroborates Theorem 2: at scarce channel budgets, the optimal selector maximizes the geometric volume of  $\mathcal{F}_S$ , not individual channel energy. Energy-based selection favors high-variance channels often mutually correlated (all carrying the dominant background direction  $B(X)$ ), whereas CovVol explicitly penalizes such redundancy via Schur-complement updates, preferring channels whose residuals span complementary directions.

**Detailed Finding 3: OOD failure is interference breakdown.** The most important validation comes from Figure 4(c)(d), which links the geometric quantities of Section 3.2.2 to empirical failure under distribution shift. Panel (c) visualizes  $\hat{X}^{(c)}$  at multiple corruption severities: as severity grows, the foreground progressively dissolves, background bleeds back into the pixel field, and the sharp foreground silhouette characteristic of the in-distribution regime gradually disappears. This is the direct visual fingerprint of interference failure: the classifier weights  $w^{(c)}$  that enforced (B1) for clean inputs no longer satisfy the boundary condition when  $B(X)$  itself is perturbed by the corruption. Background annihilation becomes imperfect, foreground emergence weakens, classification accuracy drops.

Panel (d) makes the visual story quantitative. We plot, as a function of corruption severity  $s \in \{0, 1, \dots, 5\}$ , the normalized evolution of five quantities: classification accuracy (left axis, downward), the covariance-volume determinant  $\log \det(\Sigma)$ , the effective rank of  $\Sigma$ , the trace  $\text{Tr}(\Sigma)$ , and the feature-space MMD distance [9] between clean and corrupted distributions (right axis, upward). The MMD distance is computed via the standard Gaussian kernel estimator  $\text{MMD}^2 = \mathbb{E}_{X, X' \sim P}[K(X, X')] - 2\mathbb{E}_{X \sim P, X' \sim Q}[K(X, X')] + \mathbb{E}_{Y, Y' \sim Q}[K(Y, Y')]$  between the clean and corrupted feature distributions. All four geometric indicators of interference capacity, accuracy, log det, effective rank, and trace, decay monotonically and approximately proportionally as severity increases, while the MMD grows monotonically in tandem.

This co-evolution is strong empirical support for the destructive-interference theory. In the SFH picture, distribution shift is an abstract concept (“activations move to a different part of feature space”) with no principled link to the algebraic objects of classification. In our framework, the link is explicit: the covariance volume  $\det(\Sigma)$  governs the volume of the admissible interference family  $\mathcal{F}_S$  (Theorem 2), and its collapse under corruption directly predicts the collapse of the classifier’s capacity to enforce (B1)–(B2). Out-of-distribution failure is not vague distribution shift; at the geometric level, it is the erosion of the very covariance volume that interference-based classification requires. To our knowledge, this is the first experimental result that renders OOD generalization concrete as a linear-algebraic phenomenon in the reconstructed pixel manifold.

## C.7 CAM Comparison: Trade-off Analysis

This appendix expands Section 3.3.

**Detailed faithfulness mechanism.** The faithfulness gap, especially the order-of-magnitude separation on Deletion AUC, has a direct mechanistic explanation rooted in our theory. Grad-CAM-family methods compute a coarse-resolution saliency at a convolutional feature map (typically the last conv block, with spatial resolution  $7 \times 7$  for ConvNeXt-Base or  $14 \times 14$  for ResNet-50 at  $224 \times 224$  input) and upsample it back to pixel space via bilinear interpolation, discarding the high-frequency adjoint structure that our LAC cascade is specifically designed to recover (Section 2.3). As a consequence, the pixels they rank as “most important” are the low-spatial-frequency centroids of the object, not the pixels that the classifier actually integrates to form its decision. The zero-hallucination guarantee of Corollary 1 ensures, by contrast, that every pixel ranked nonzero by our attribution lies within the effective receptive field of a genuinely active channel, so insertion and deletion act precisely on the decision-relevant pixels. The order-of-magnitude gap on Deletion AUC is the empirical signature of this structural fidelity.

**Why we dominate mAP and Pointing.** Both metrics are threshold-free and measure whether the attribution places its peak energy on the discriminative region. Pointing simply asks whether the single highest-scoring pixel lies inside the foreground mask; mAP integrates a ranking-based precision across all thresholds. Under our framework, Corollary 1 guarantees that every nonzero spatial variation of  $\hat{X}^{(c)}$  lies within the dilated union of active effective receptive fields, and Theorem 3

shows that the energetic weight on those pixels is monotone in the classifier’s decision signal. The peak and the rank ordering are therefore, by construction, concentrated on the pixels the classifier actually uses, which is precisely what these two metrics reward.

**Why mIoU penalizes us.** mIoU, by contrast, measures the overlap between a binarized saliency map and the full foreground mask. CAM-family methods, which upsample from a coarse feature-map resolution, naturally produce smooth, plateau-shaped maps that fill out the entire object region at uniform magnitude, ideal for mIoU even though this plateau does not reflect any underlying model-internal gradient. Our reconstruction, by design, places energy in proportion to the interference-weighted structural contribution of each pixel (Section 3.2.2), which is concentrated on object-discriminative sub-regions (e.g., the face and limbs of a pet, the head and wings of a bird) rather than uniformly distributed across the full silhouette. A metric that rewards uniform coverage will systematically favor the smoother, less informative map. The same logic explains the pAcc pattern on CUB, where birds occupy a small fraction of the image and bulk-overlap metrics are especially threshold-sensitive.

**Why the trade-off is consistent with our theory.** The destructive-interference account predicts that classification is driven by sparse, sub-object residuals  $\delta_i(X)$ , not by uniform enhancement of the entire foreground silhouette. An attribution map that is faithful to this theory should therefore be energetically concentrated on sub-object discriminative regions, scoring highly on peak-localization metrics (mAP, Pointing) and on faithfulness (Ins/Del), but necessarily lower on bulk-overlap metrics (mIoU) that implicitly presuppose uniform object-level attention. Score-CAM’s strong mIoU is consistent with this reading: its forward-pass score-weighted aggregation produces a smoothed region-level map that aligns well with the full mask but, as Table 2 shows, loses its connection to the decision mechanism (Deletion AUC 0.4267 on CUB,  $2.6\times$  ours). The split between faithfulness and bulk-localization is not an inconsistency but the expected signature of attribution that is mechanistically faithful to an interference-based classifier rather than to an “object saliency” heuristic.

## C.8 Causal ECR Ablation: Full Protocol and Analysis

This appendix expands Section 3.4.

**ECR as an H1-derived concept localizer.** Under H1, every channel inversion decomposes as  $\tilde{V}_{L-1,i}(X) = C_i(X)B(X) + \delta_i(X)$ . Substituting into the ECR definition,

$$\text{ECR}_i = \frac{\sum_{(u,v) \in \mathcal{R}} \sum_k [C_i B + \delta_i]_k(u, v)^2}{\sum_{(u,v)} \sum_k [C_i B + \delta_i]_k(u, v)^2}. \quad (107)$$

Since  $B(X_{\text{ref}})$  is the same scene-level direction shared by every channel, its projection onto the region  $\mathcal{R}$  contributes a near-constant term across all  $i$  (modulated only by the per-channel coefficient  $C_i$ ). The cross-channel variation of  $\text{ECR}_i$  is therefore dominated by how strongly the residual  $\delta_i$  localizes inside  $\mathcal{R}$ . A channel with high  $\text{ECR}_i$  is, by this measure, a candidate carrier of the concept whose spatial extent is  $\mathcal{R}$ .

**Detailed experimental protocol.** A single Birman image  $X_{\text{ref}}$  is selected from the test set, and its tail region  $\mathcal{R}$  is manually annotated. From this single annotation we compute  $\{\text{ECR}_i\}_{i=1}^{C_{L-1}}$  and freeze a ranking of all channels. Three ablation orders are then defined: **Descending** (high ECR first; tail-encoding channels removed first), **Ascending** (low ECR first; tail-irrelevant channels removed first), **Random** (uniform random order, mean $\pm$ std over 50 seeds). The frozen ranking obtained from  $X_{\text{ref}}$  alone is then applied to every Birman image in the test set: for each target image  $X$  and pruning fraction  $\rho \in [0, 1]$ , we zero the top- $\lceil \rho C_{L-1} \rceil$  channels of  $h(X)$  in the chosen order, re-run the classifier head, and record  $P(\text{Birman} \mid X, \rho)$ , averaging across all Birman test images.

**Detailed numerical observations.** The Descending curve collapses substantially faster than the random baseline at every ablation rate. At 40% ablation,  $P(\text{Birman})$  has fallen to roughly 0.84 (Descending), compared with approximately 0.90 (Random) and 0.94 (Ascending). At 70% the gap widens to 0.61 vs. 0.83 vs. 0.88. Because the only information used to order the channels is the tail mask on a single held-out image, this rapid decay is direct causal evidence that the high-ECR channels genuinely participate in the classifier’s decision for the Birman class.

The Ascending curve exhibits a mirror-image pattern. It tracks the full-model baseline almost exactly up to  $\sim 60\%$  ablation and even lies above the Random baseline throughout the mid-ablation regime, removing the least-tail-relevant channels does not merely fail to hurt the prediction, it mildly improves it, presumably because the retained channels are enriched for the target concept. A marked drop occurs only beyond  $\sim 75\%$  ablation, where the remaining budget is insufficient to maintain the destructive-interference geometry regardless of concept selectivity. At 100% ablation all three curves converge to chance.

The separation between Descending and Ascending, exceeding 0.3 in  $P(\text{Birman})$  across a wide ablation range, is the central empirical result. It would be impossible if the channels were either spatially unspecialized (both orders should match Random) or sparsified onto object regions in the SFH sense (both orders should collapse quickly). The observed ordering is the precise signature predicted by our theory: channels are holographically full (H1), but their residuals  $\delta_i$  carry identifiable sub-object concepts, and these residuals are causally load-bearing for classification.

**Why a single-image mask generalizes.** Under H1, the shared background  $B(X)$  varies with  $X$ , but the channel identity that encodes each sub-concept does not: channel  $i$  is the same linear filter for every input, and its residual  $\delta_i(X)$  is the same semantic probe evaluated on different scenes. A region mask on a single image therefore surfaces channels whose residuals localize on that region; because the channel index transfers across images of the same class, the ranking transfers as well. This is the pixel-space analog of the feature universality empirically observed in mechanistic interpretability of language models [5], and our inversion operator renders it directly visible and directly testable.

**Implications.** Three implications follow. First, this is the first demonstration that a hallucination-free pixel-space inversion enables causal concept localization at the channel level from a single spatial annotation, a capability unavailable to any CAM-family method, which produces a per-sample heatmap but no channel-level semantic handle. Second, the experiment provides empirical evidence for the destructive-interference account: the classifier’s behavior is not merely correlated with channel residuals (as an attribution method might detect), it is causally gated by them in precisely the order predicted by ECR. Third, the combination of ECR and ablation yields a simple, supervision-light procedure for discovering and verifying concept-selective channels in any frozen CNN classifier, with potential applications in model auditing and concept-level debugging.

## C.9 ViT Generalization: Detailed Architecture and Visualization Analysis

This appendix expands Section 3.5.

**Hybrid architecture details.** The frozen ConvNeXt-Base encoder produces  $\mathbf{h}_{L-1}(X) \in \mathbb{R}^{1024 \times 7 \times 7}$  at the deepest stage. This tensor is spatially flattened and transposed into a sequence of 49 tokens of dimension 1024. The token sequence is then processed by three stacked multi-head self-attention blocks  $\mathcal{A}_1, \mathcal{A}_2, \mathcal{A}_3$  following the standard Transformer encoder design (multi-head self-attention with LayerNorm and residual connections), followed by a linear classification head. During training only the attention blocks and the classification head are optimized; the entire convolutional encoder is held frozen at its ImageNet-pretrained weights. After convergence the LAC cascade trained per Section 2.4 is loaded unchanged and applied for visualization, with no fine-tuning.

**Adjoint composition: full algebraic discussion.** The visualization protocol exploits the fact that the LAC cascade is a well-defined adjoint operator from the deepest feature manifold  $\mathbf{h}_{L-1}$  back to the pixel space. For any scalar  $Q$  defined on the attention pathway, the gradient  $\partial Q / \partial \mathbf{h}_{L-1}$  is a single-step VJP through the attention stack, computable in standard automatic differentiation. Feeding this gradient as the seed of the LAC cascade yields

$$\tilde{V}_Q(X) = \Psi_{\text{LAC}_{\text{stem}}} \circ J_{\text{stem}}^T \circ \Psi_{\text{LAC}_0} \circ J_0^T \circ \cdots \circ \Psi_{\text{LAC}_{L-1}} \circ J_{L-1}^T (\partial Q / \partial \mathbf{h}_{L-1}). \quad (108)$$

All structural guarantees that the LAC enjoys on the manifold  $\mathbf{h}_{L-1}$  transfer verbatim: Theorem 1, Corollary 1, and Proposition 3 all apply. The attention blocks are free to implement arbitrary non-convolutional operations, softmax attention, layer normalization, residual connections, even cross-attention to additional inputs, without compromising the zero-hallucination guarantee, because those operations affect only the construction of the seed, not the adjoint inversion below it.

1343 **Per-layer and per-token specializations.** For a layer-wise visualization at attention block  $\ell$ , we  
 1344 take  $\mathcal{Q}_\ell = \frac{1}{2} \|\mathcal{A}_\ell(\mathbf{T}(X))\|_F^2$ , the block’s total activation energy. The resulting  $\tilde{V}_{\mathcal{Q}_\ell}$  depicts the  
 1345 pixel regions whose perturbation most strongly alters the block’s overall response. For a per-token  
 1346 visualization at block  $\ell$  and token index  $t \in \{1, \dots, 49\}$ , we take  $\mathcal{Q}_{\ell,t} = \frac{1}{2} \|\mathcal{A}_\ell(\mathbf{T}(X))_{t,:}\|_2^2$ , isolating  
 1347 the spatial evidence the block relies on when computing the representation at that specific spatial  
 1348 token. Both specializations require only a single VJP at the attention side, with no LAC modification.

1349 **Detailed visual analysis.** Figure 6 shows representative outputs. Two qualitative findings emerge.  
 1350 First, the layer-wise aggregate visualizations remain coherent and photometrically faithful across all  
 1351 three attention depths. Despite the signal passing through additional non-convolutional computation  
 1352 (softmax attention plus residual MLP plus LayerNorm) before entering the LAC cascade, the pixel-  
 1353 space output preserves the geometric integrity of the input scene with no observable hallucinations.  
 1354 This is consistent with Corollary 1 applying to the seed regardless of how it was constructed upstream:  
 1355 the spatial gradient support of the visualization is bounded by the encoder’s effective receptive fields,  
 1356 irrespective of whether the seed came from a linear classifier, a self-attention block, or any other  
 1357 differentiable head.

1358 Second, the per-token visualizations exhibit a structurally meaningful progression across depth. Early  
 1359 attention tokens (column b,  $\mathcal{A}_1$ ) recover locally diffuse pixel evidence: each token’s visualization  
 1360 spans a moderately wide spatial region centered roughly on the token’s spatial location, indicating that  
 1361 early attention has not yet substantially refined the spatial selectivity inherited from the convolutional  
 1362 features. Deeper attention tokens (column d,  $\mathcal{A}_3$ ) concentrate on increasingly specific sub-object  
 1363 regions, reflecting the hierarchical refinement of attention weights across layers. Adjacent tokens  
 1364 attend to neighboring pixel regions, providing an immediate sanity check that the attention blocks  
 1365 have learned a sensible spatial geometry on top of the frozen convolutional features.

1366 **Universality.** This experiment establishes that the LAC cascade functions as a universal adjoint  
 1367 inverter for any frozen convolutional encoder: given any differentiable downstream module  $\mathcal{D}$   
 1368 operating on the deepest feature manifold, any scalar readout of  $\mathcal{D}$  can be visualized in pixel space  
 1369 by a single application of the cascade composition, with the full theoretical guarantees transferring  
 1370 intact. The LAC is trained once against the encoder’s adjoint structure and thereafter serves every  
 1371 downstream architecture: linear classifiers (Sections 3.2.1–3.4), attention stacks (this section), and in  
 1372 principle any other differentiable head. This is in sharp contrast to gradient-based attribution methods  
 1373 such as Grad-CAM, which are defined relative to a specific classifier output and must be re-derived  
 1374 whenever the head architecture changes. From the perspective of our theory, the universality is not  
 1375 surprising but structural: the covariance-volume geometry and destructive-interference mechanism  
 1376 of Section 3.2.2 are properties of the feature manifold (i.e., of the encoder), and any downstream  
 1377 architecture that consumes that manifold inherits access to the same interference substrate through  
 1378 the shared adjoint.

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